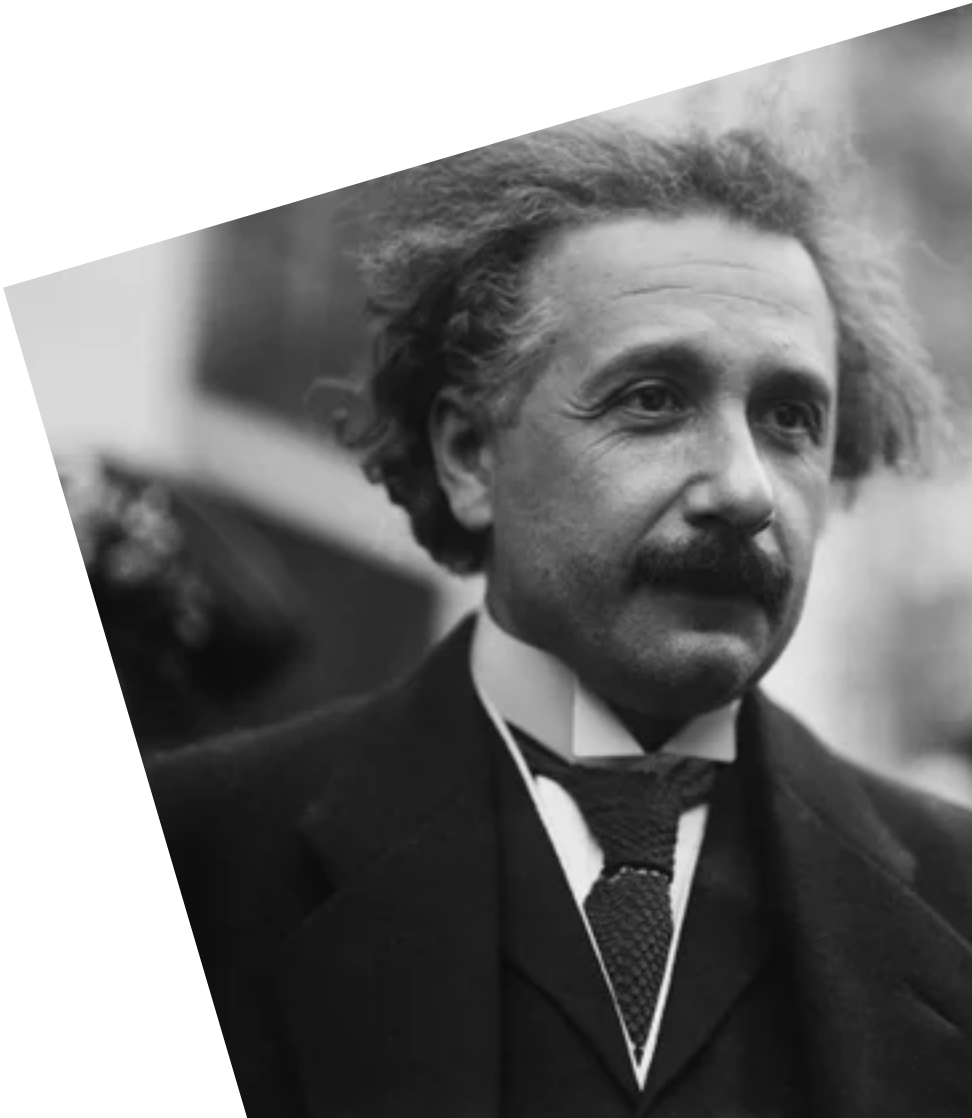
PHYC10009 Physics Foundations



Modern Physics: Photons
Week 10 Lecture 1

Professor Phillip Urquijo
School of Physics
University of Melbourne

27.3 Packets of electromagnetic energy: photons

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The beginnings of ***Quantum Physics***

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$$E = hf$$

E is the photon energy (in J)
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$h = 6.63 \times 10^{-34}\ \mathrm{J\,.\,s}$
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Thus we don't normally observe the discrete nature of light

Packets of electromagnetic energy: photons

Example problem

How many photons are emitted each second by a $1\mathrm{mW}$ laser pointer ($\lambda = 635\mathrm{nm}$) ?

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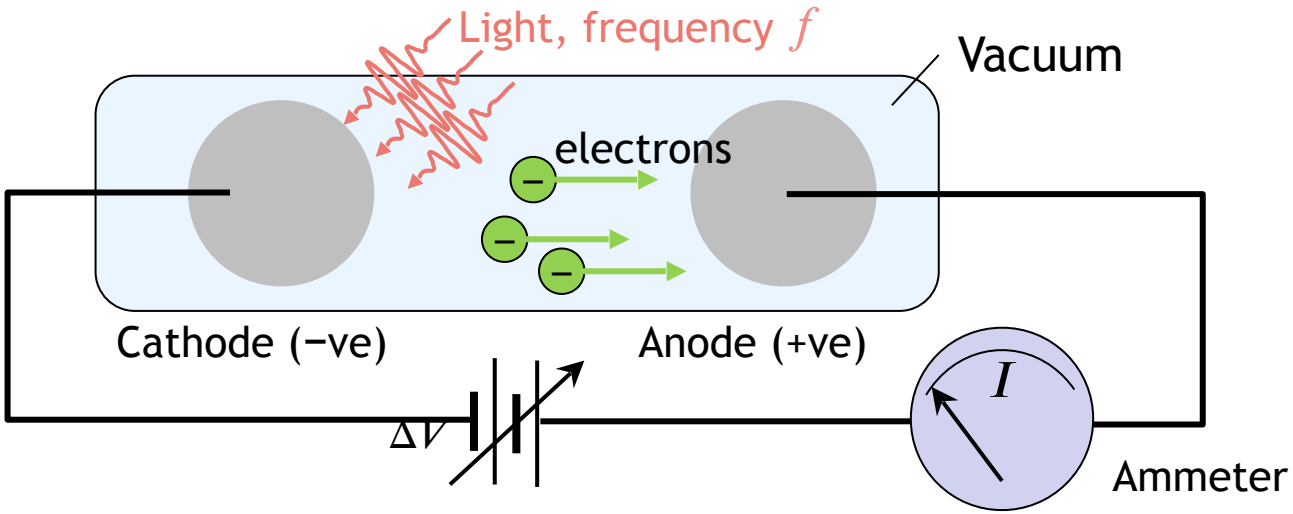
That's a lot: we don't "see" individual photons (though it ***is*** possible for human eye to see individual photons, if you are patient)

27.3 Photoelectric effect

- EM radiation is emitted and absorbed in discrete packets (photons)



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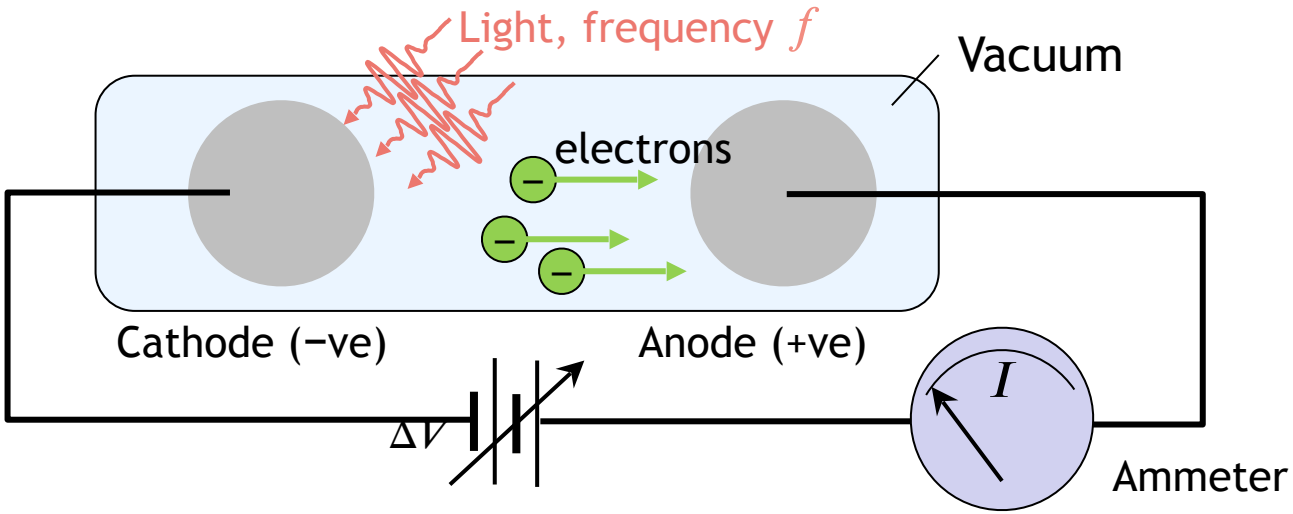

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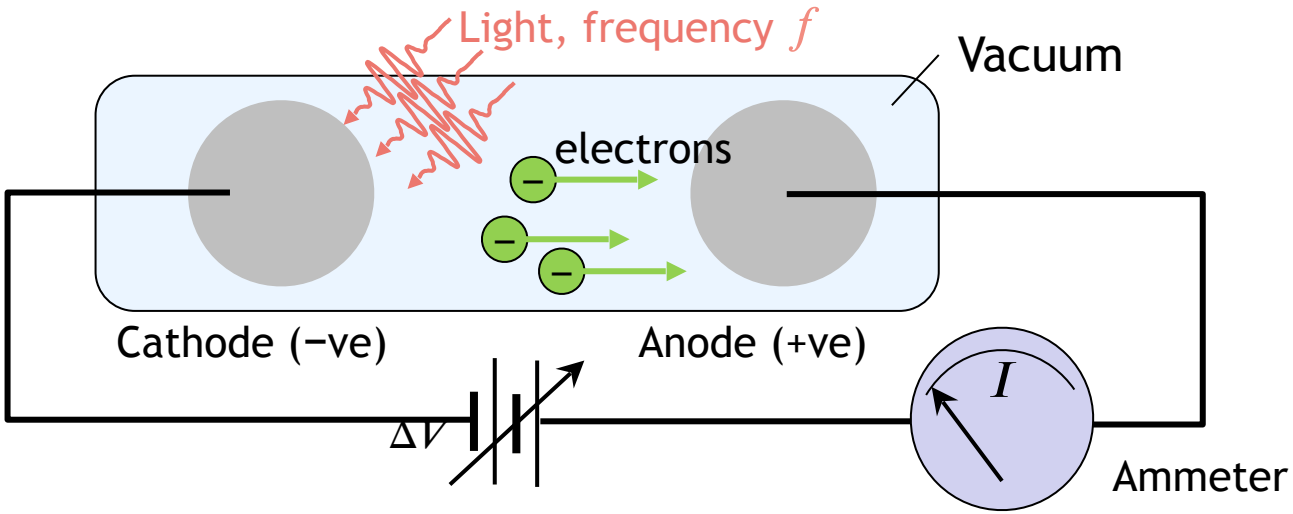

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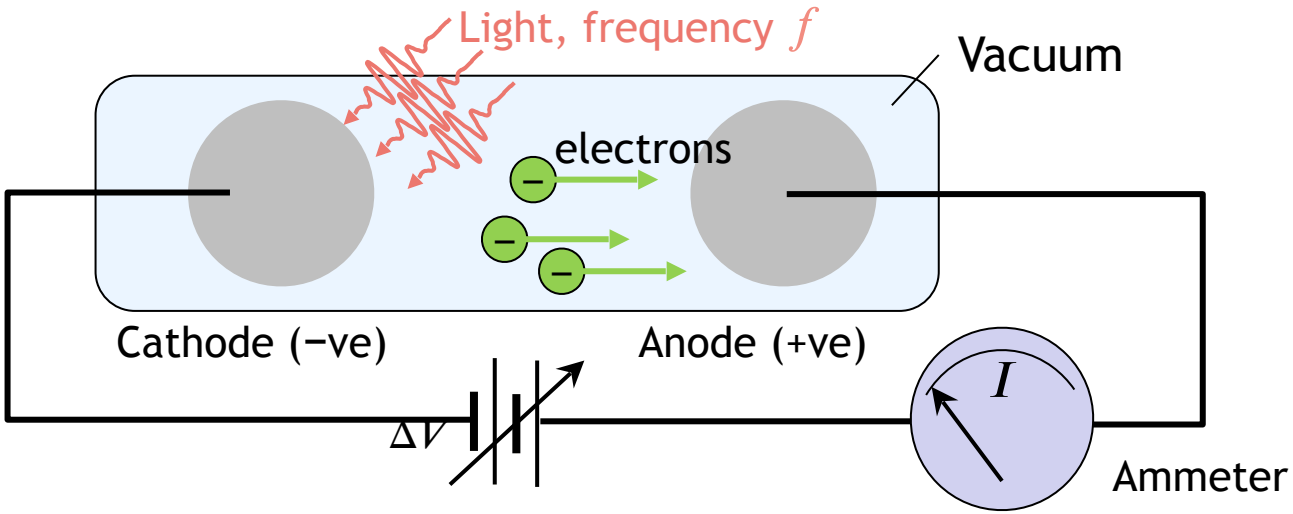
- Light on metal surface (in vacuum) → electrons emitted
- Use variable voltage source V, measure current I



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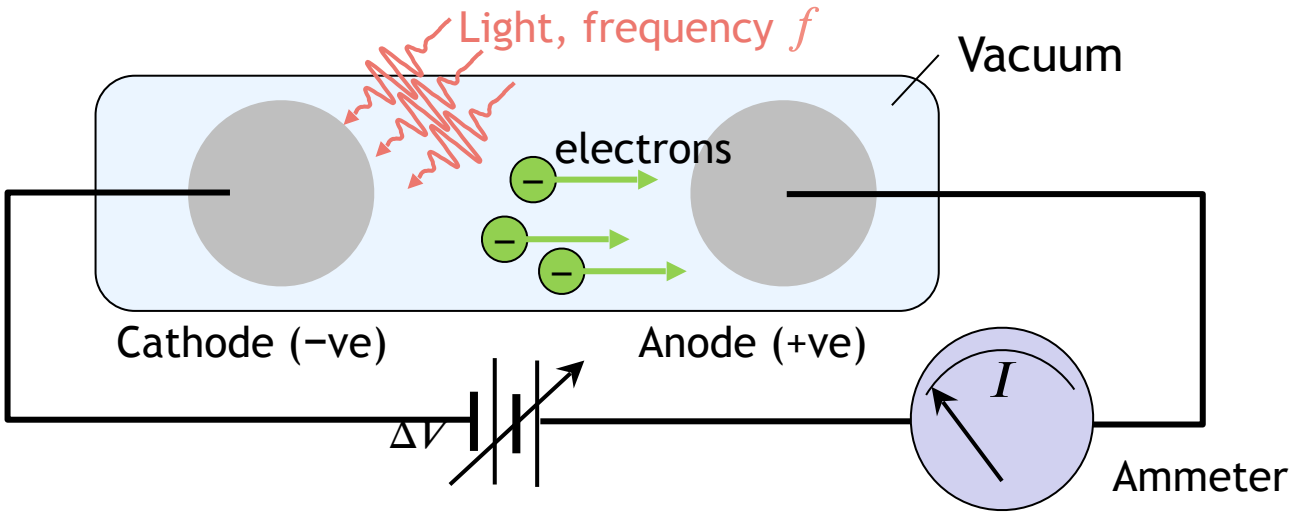
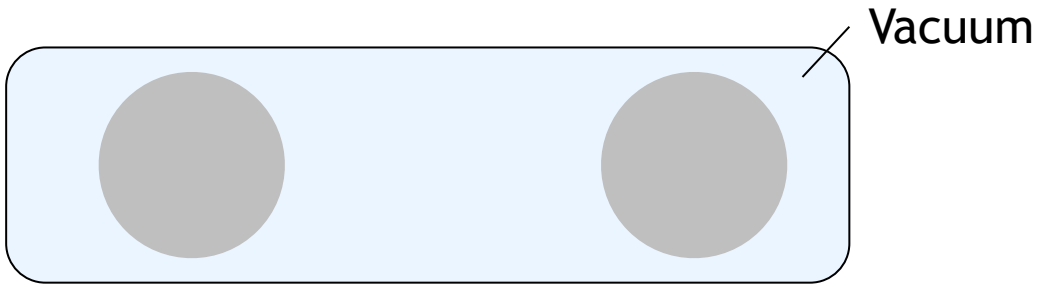


Photoelectric effect: the experiment

- Illuminate with light



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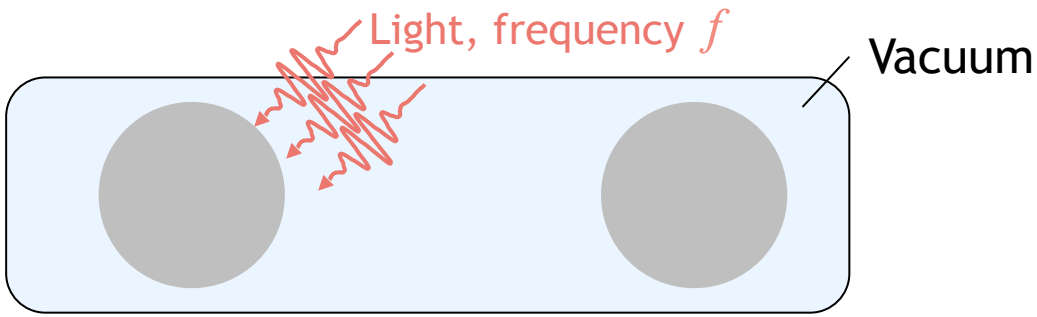

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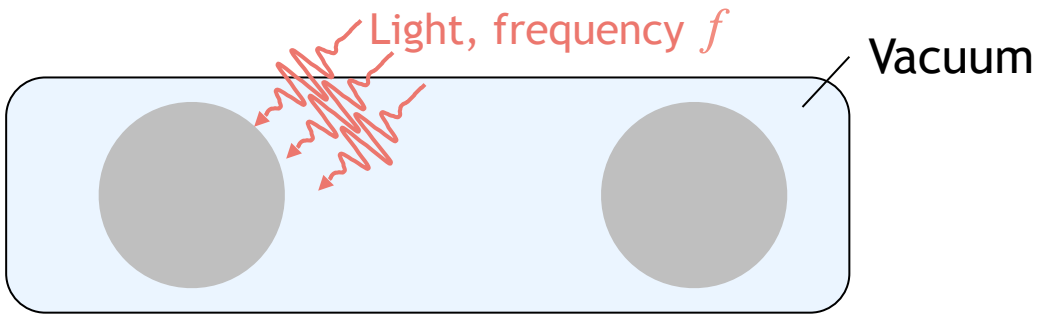

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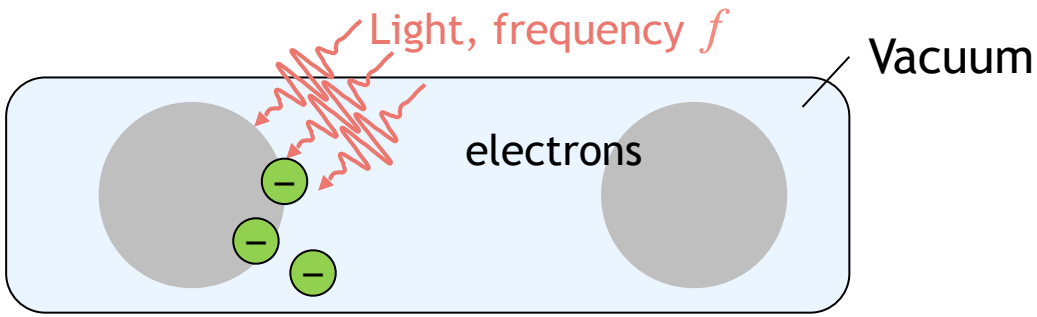

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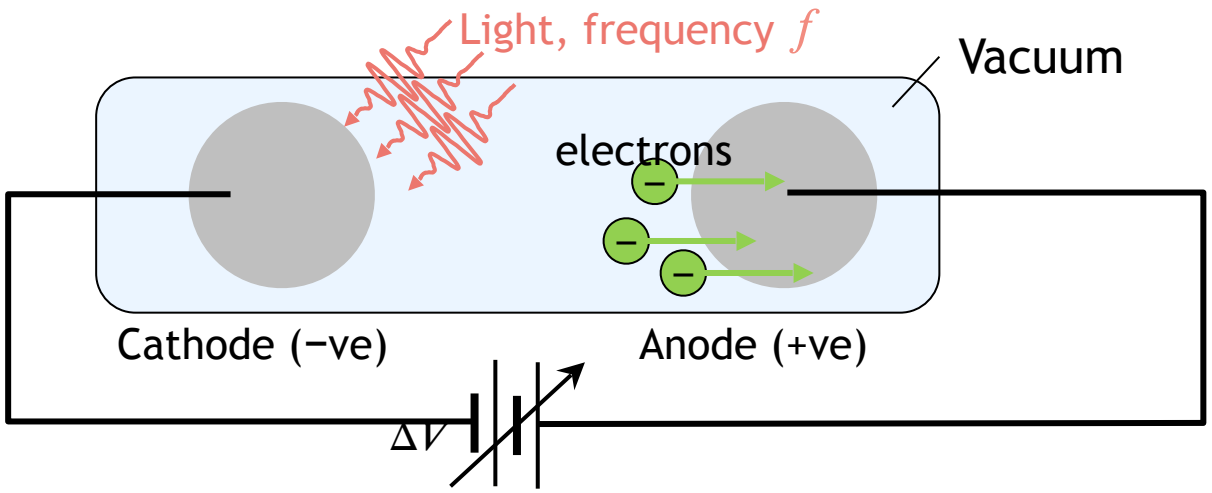

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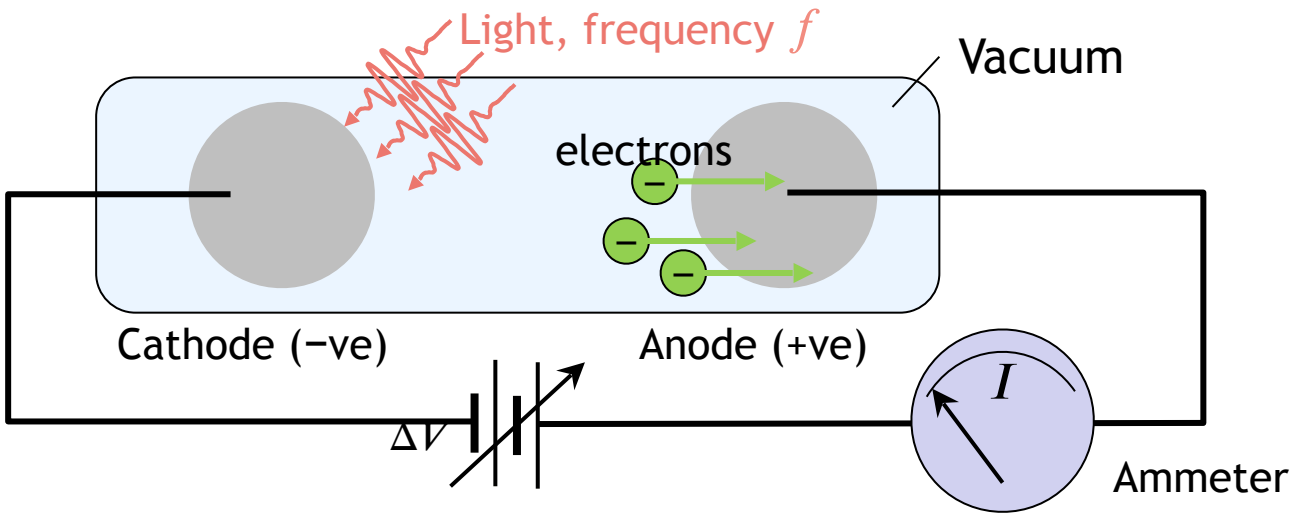

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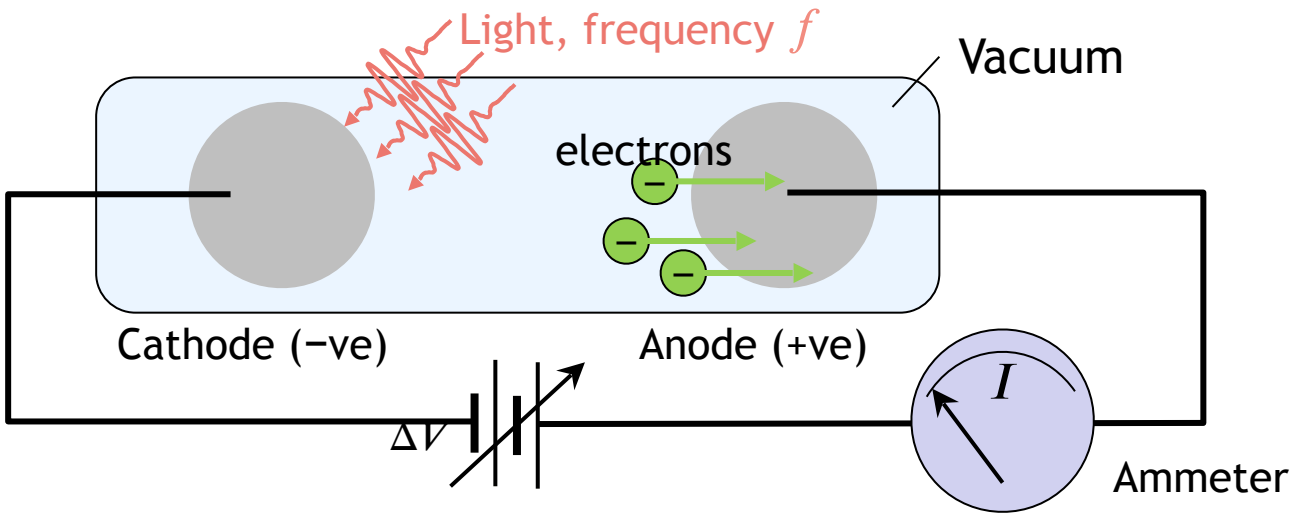
NB: Electrons ejected by LIGHT - not because of applied voltage



Photoelectric effect

1. Current is proportional to *intensity*
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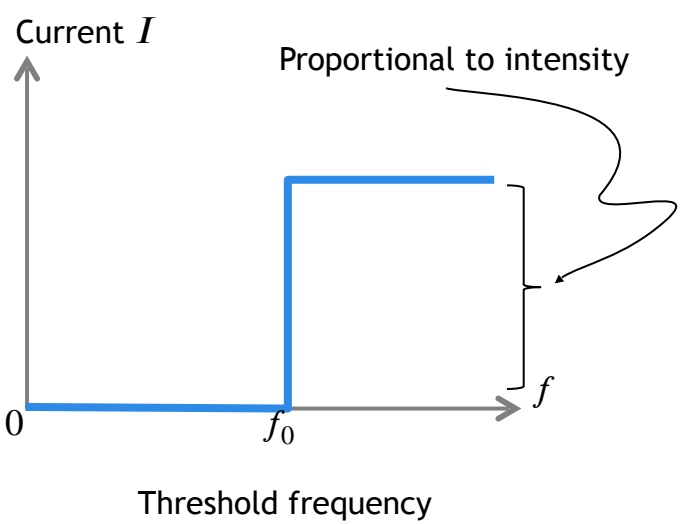

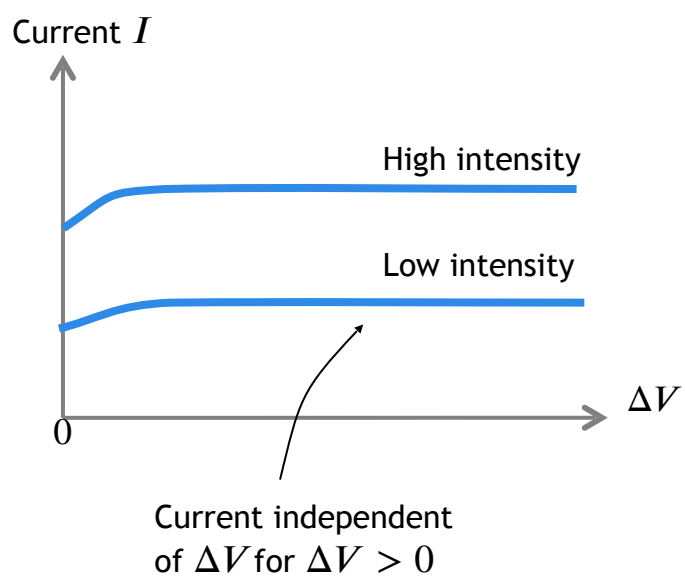

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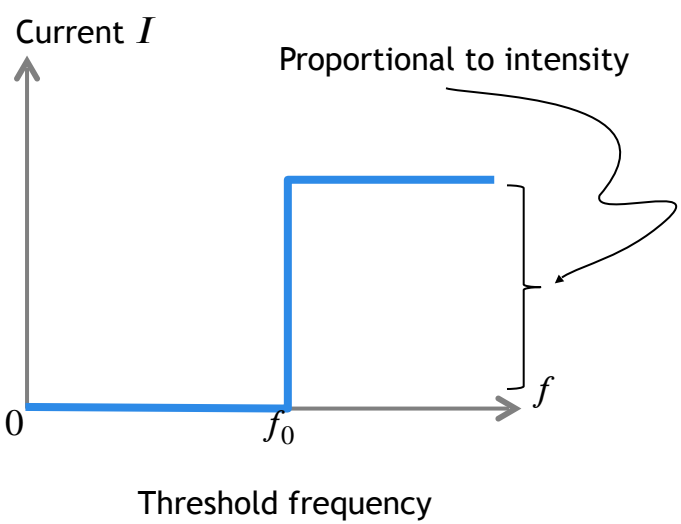

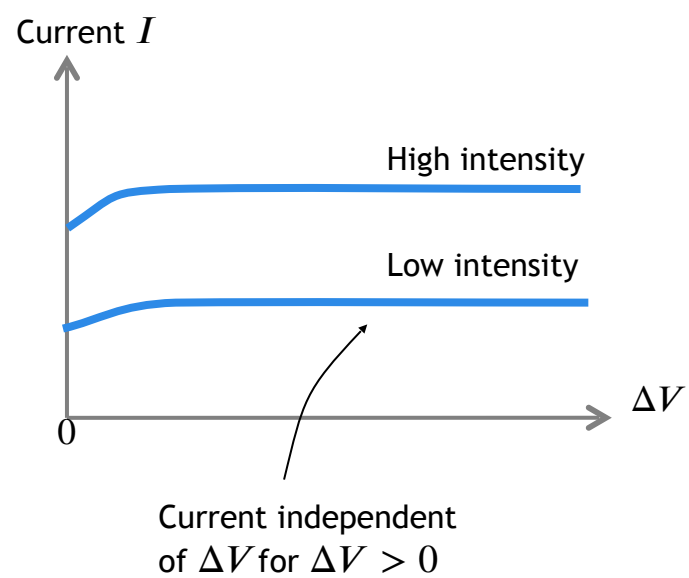

D Na-14 Photoelectric effect

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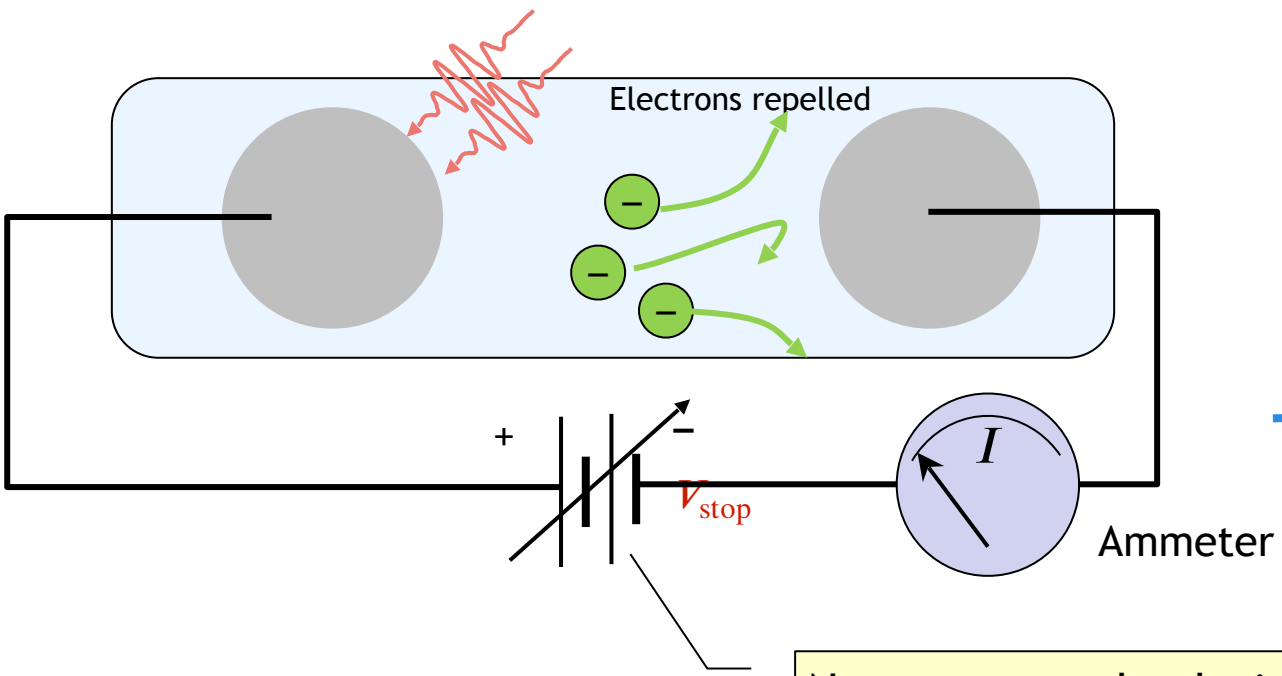

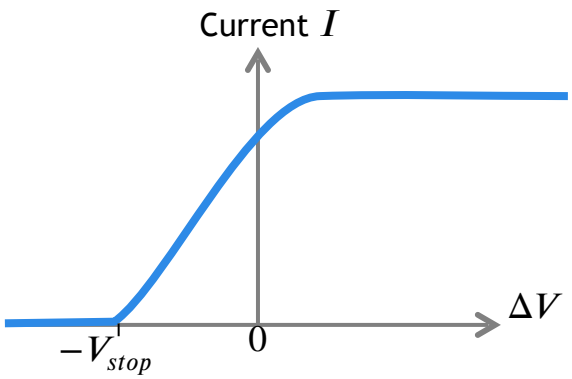

Note: reversed polarity

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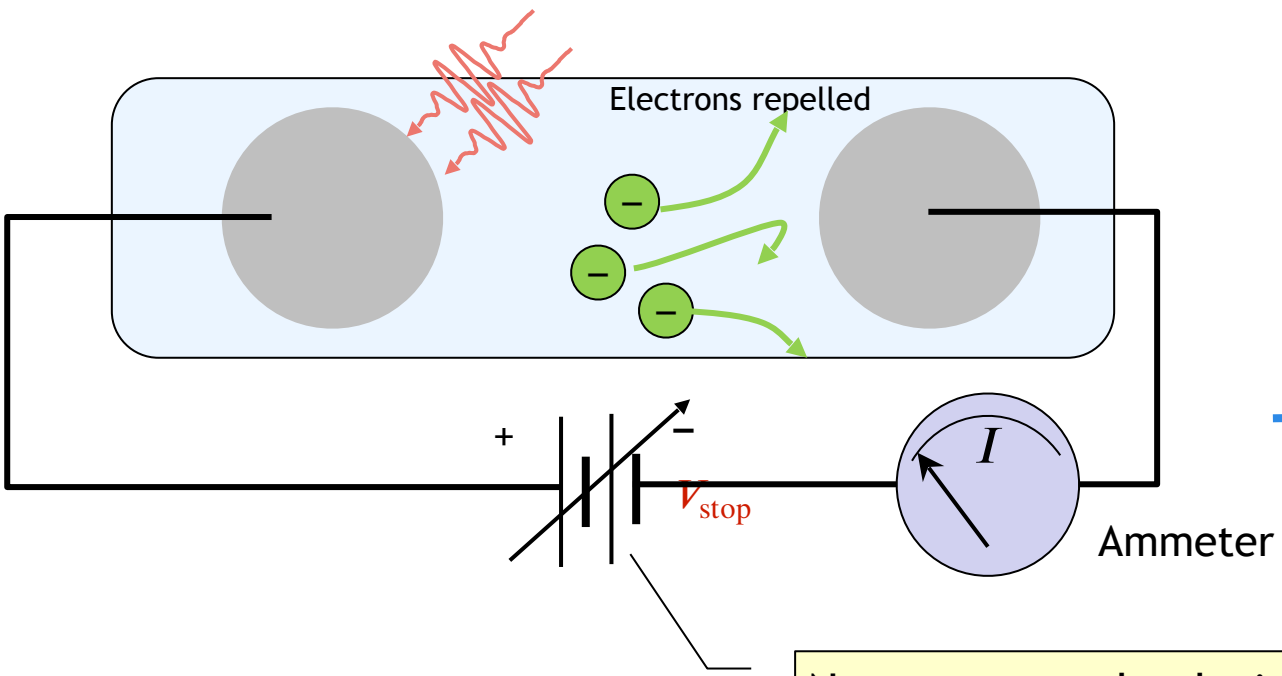

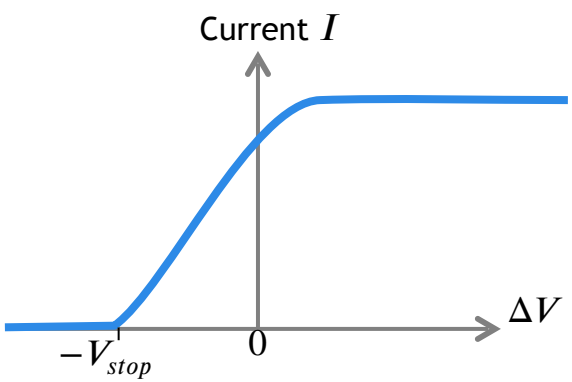

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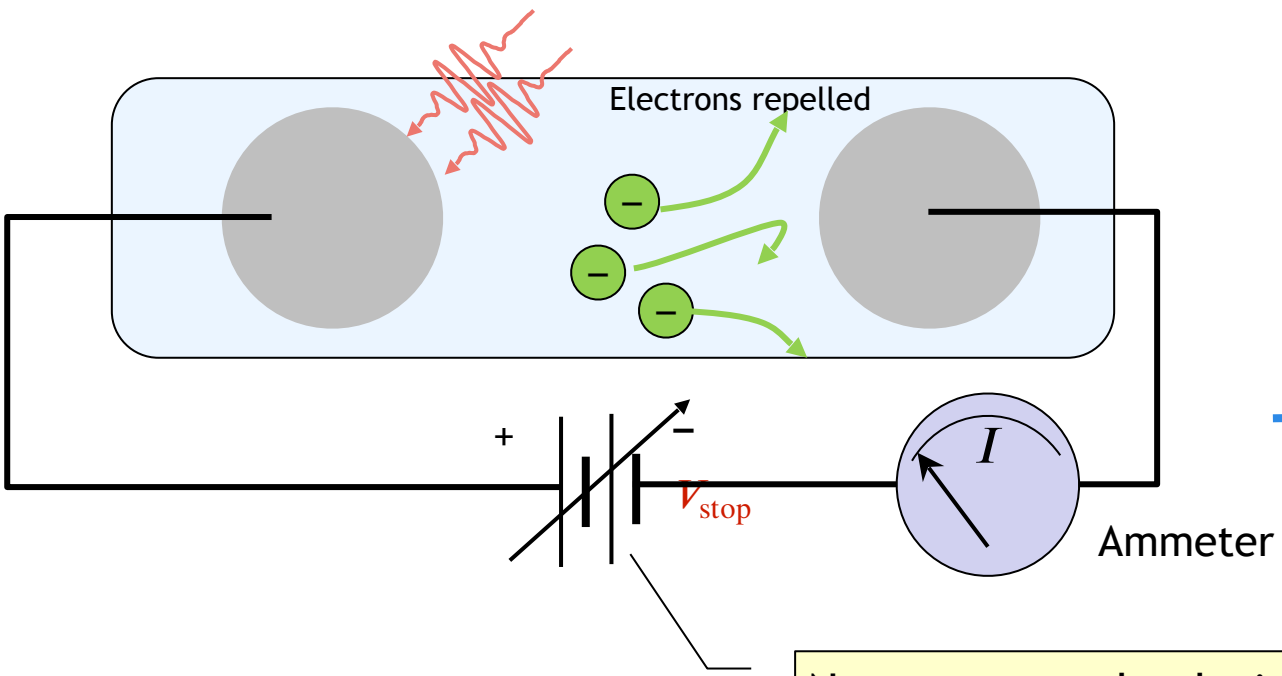

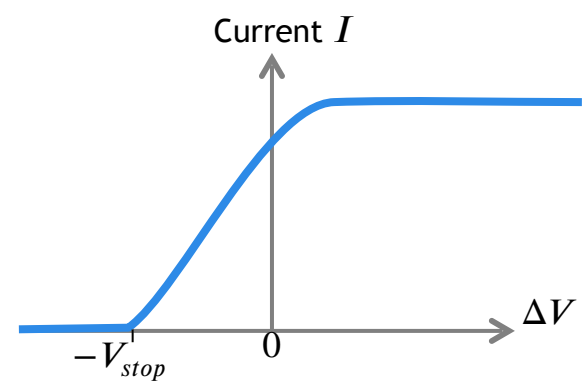

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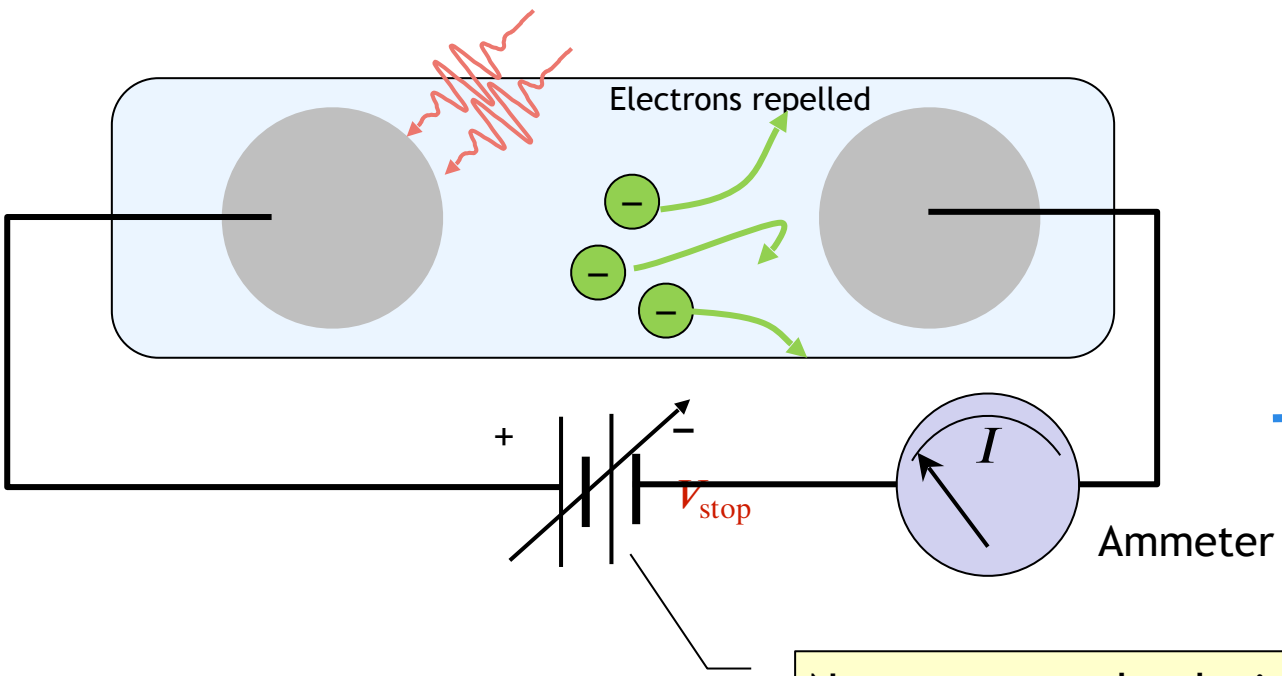

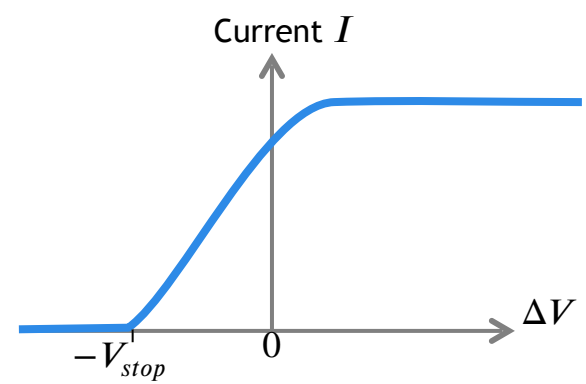

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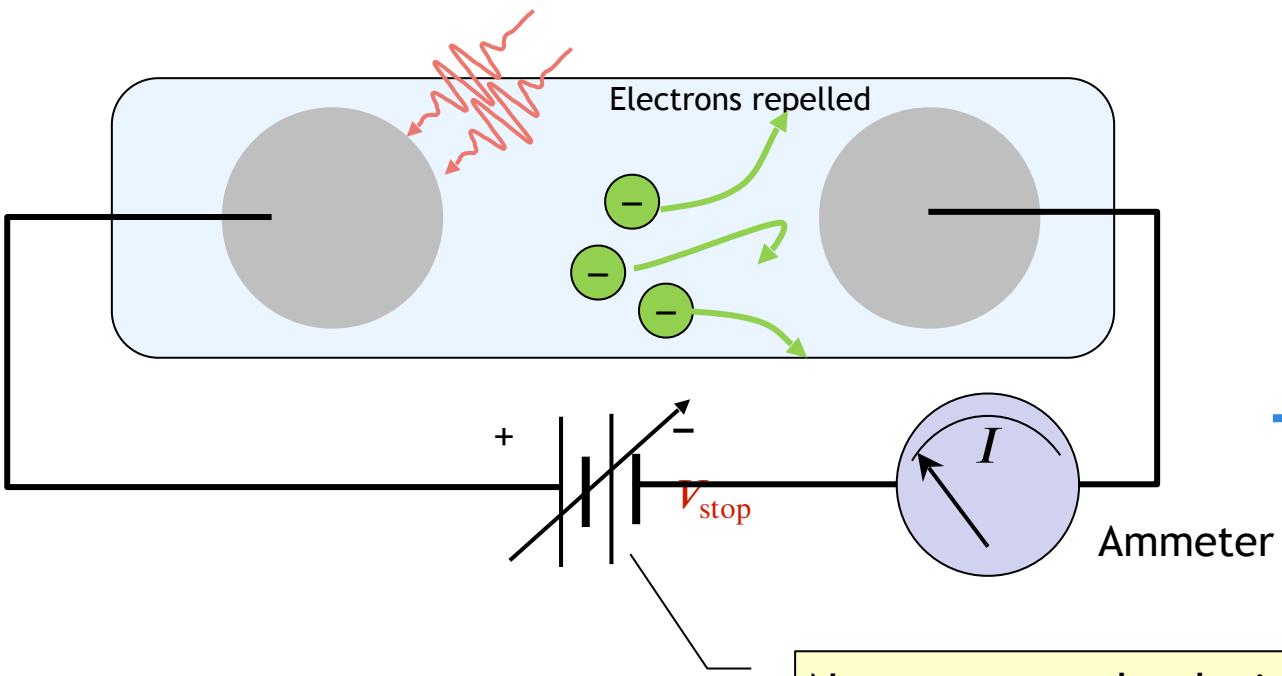

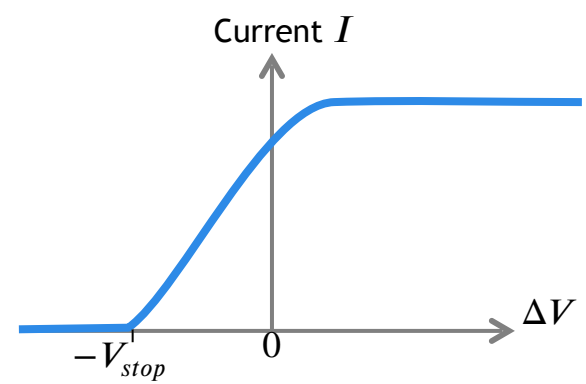

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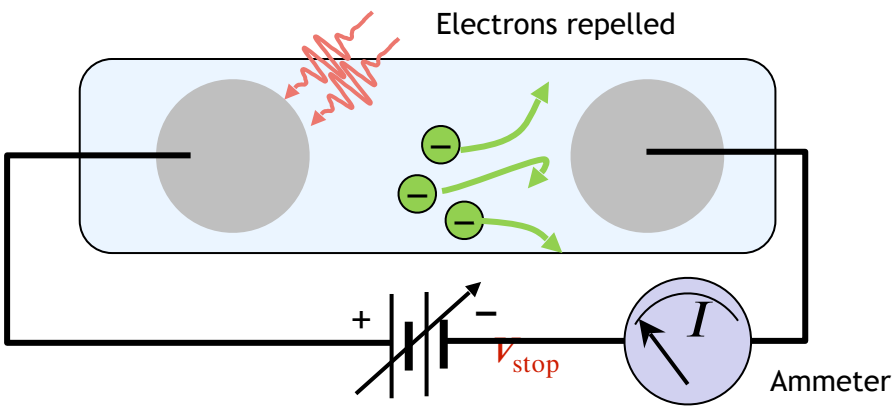
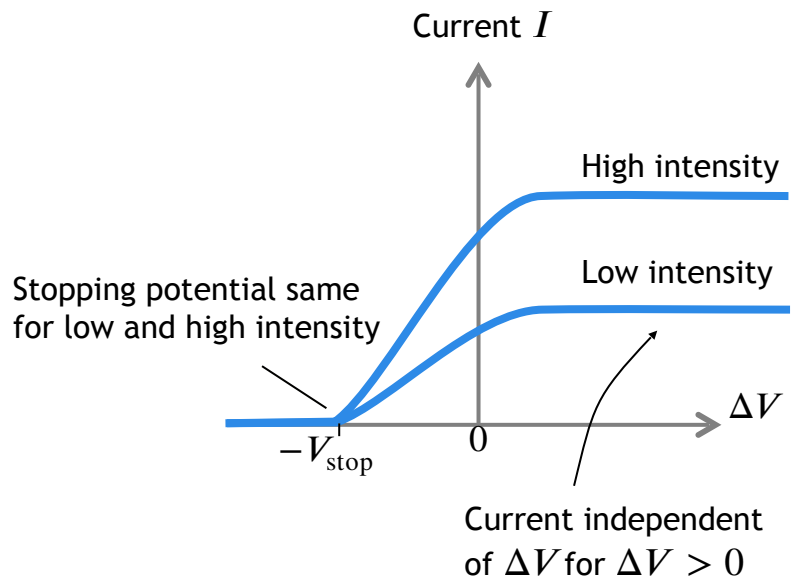
Photoelectric effect

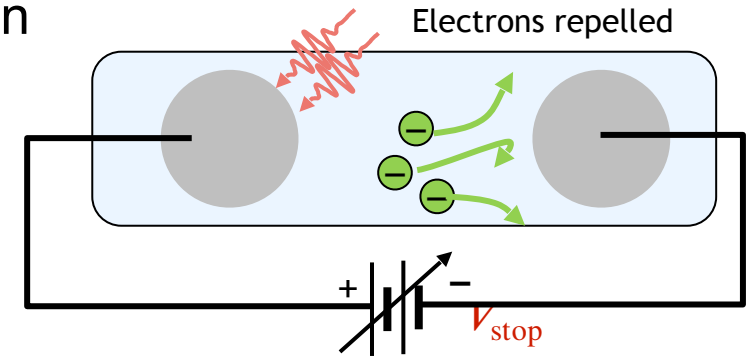
3. Current stopped by applying some minimum *stopping potential* ΔV_{stop}





Stopping potential

- Negative voltage on anode = repulsion



- Stopping potential *vs* frequency

0

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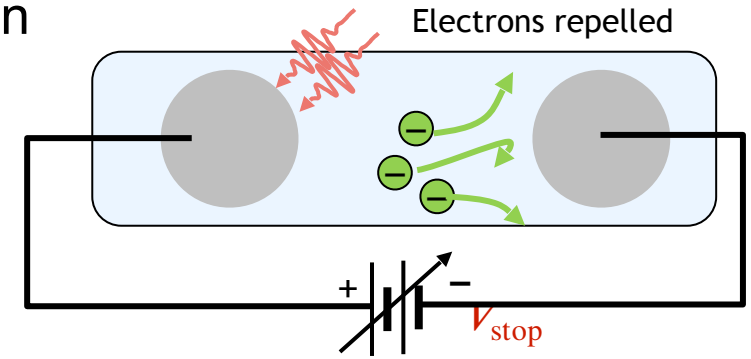

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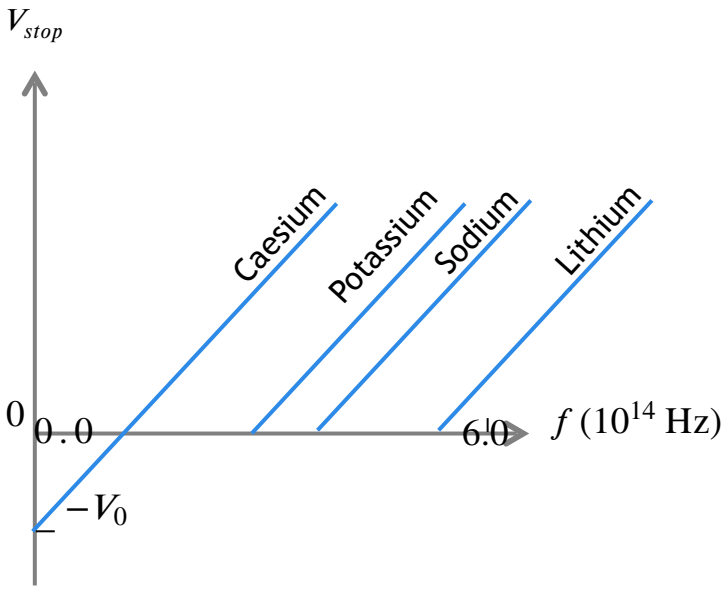

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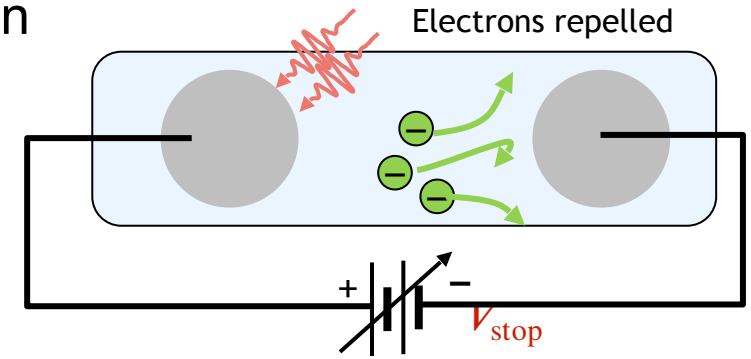
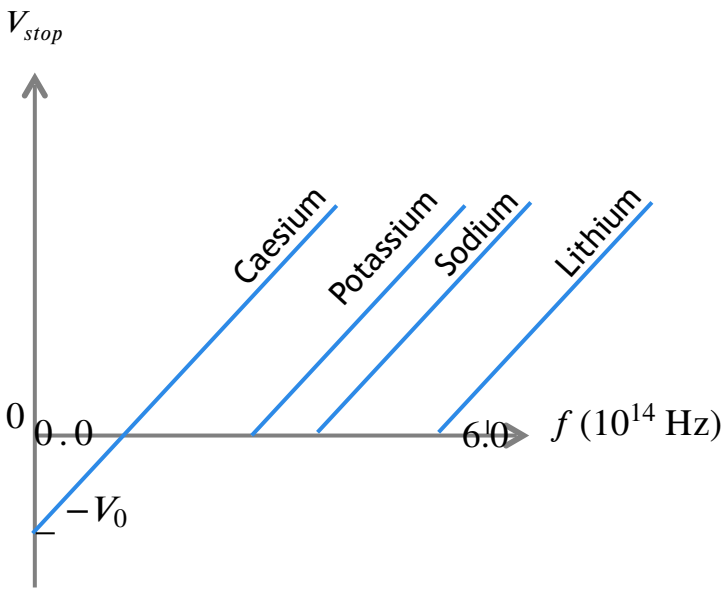


- Stopping potential *vs* frequency

$$eV_{\text{stop}} = hf - W_0$$



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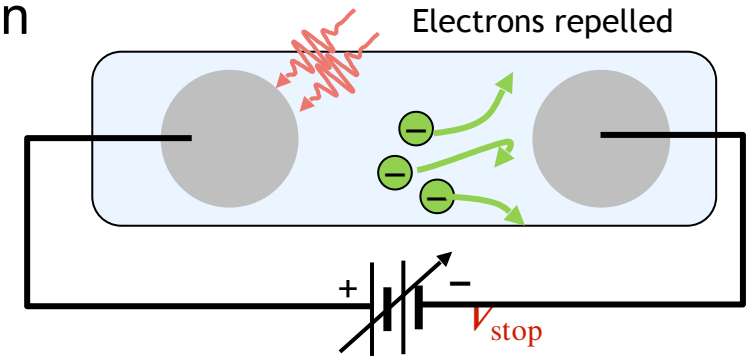

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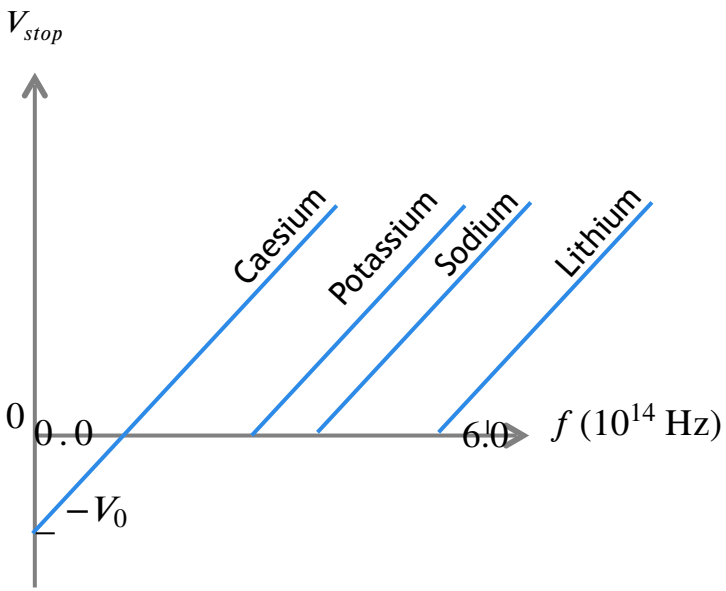

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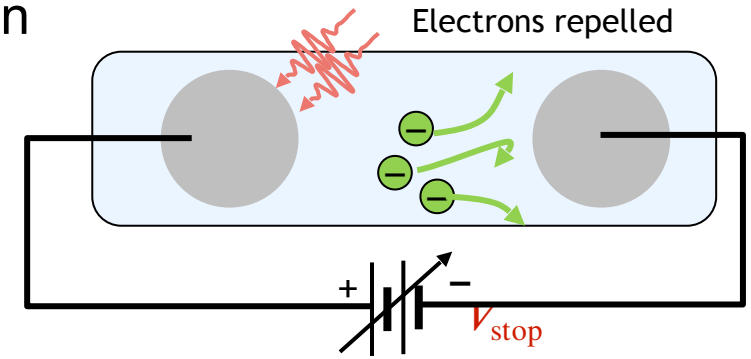

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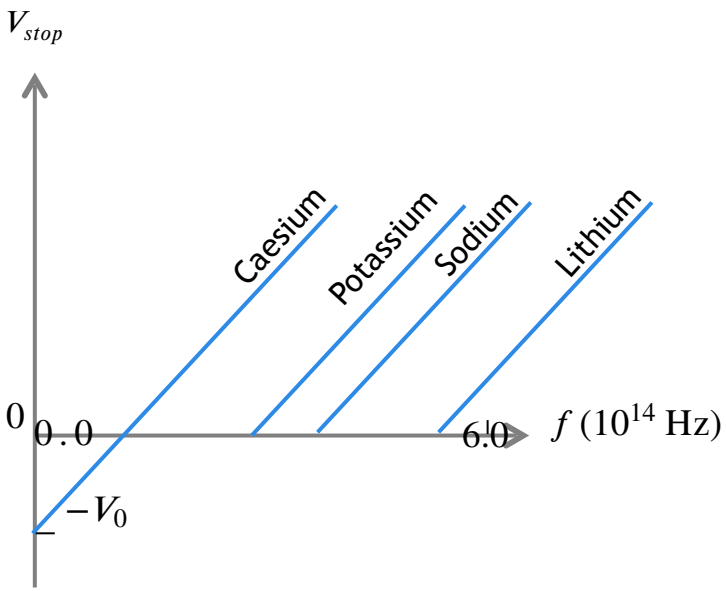

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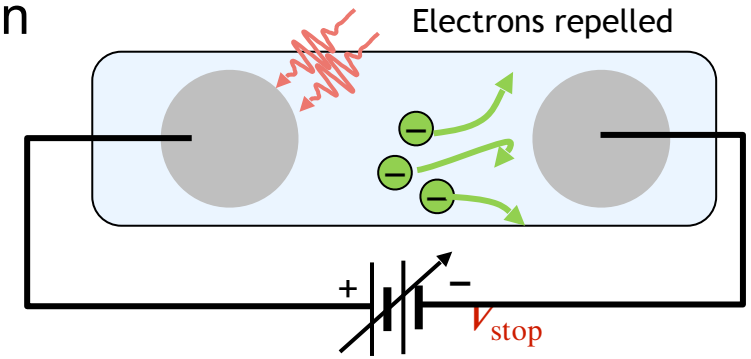
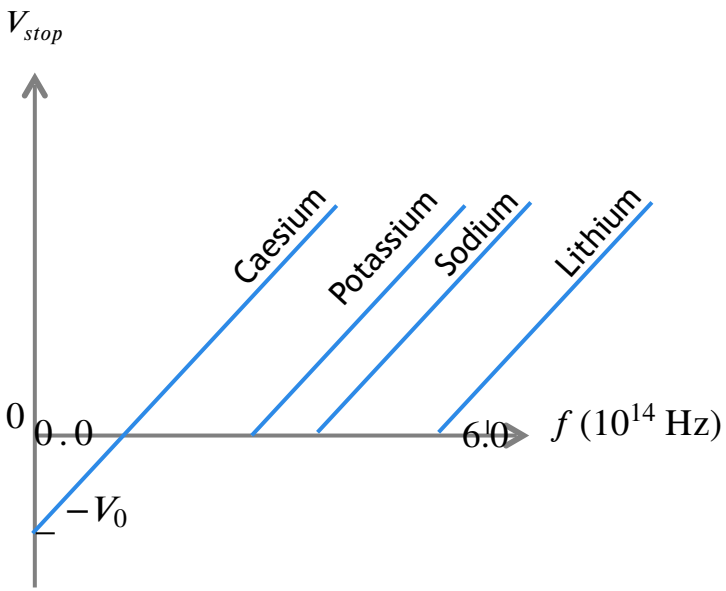


- Stopping potential *vs* frequency

$$eV_{\text{stop}} = hf - W_0$$

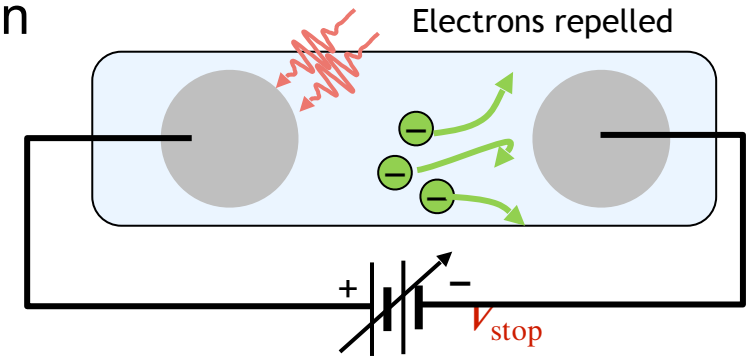
$$V_{\text{stop}} = \frac{h}{e}f - V_0$$



- Slope is constant

Stopping potential

- Negative voltage on anode = repulsion



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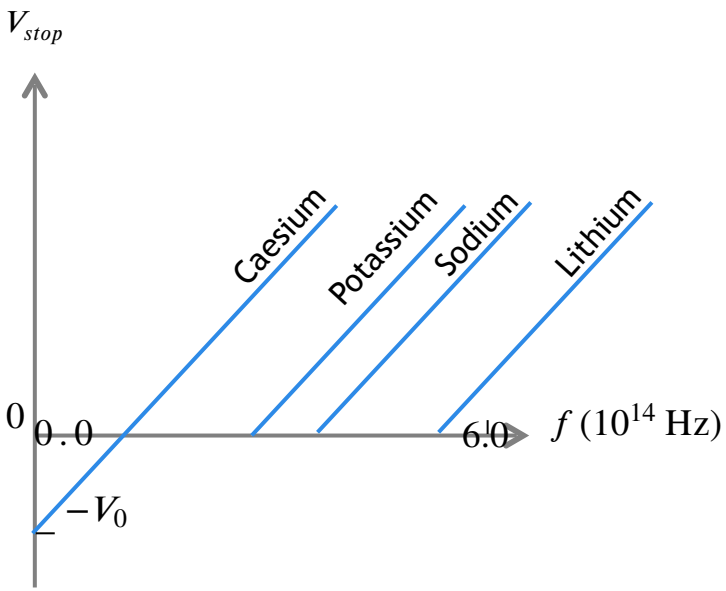
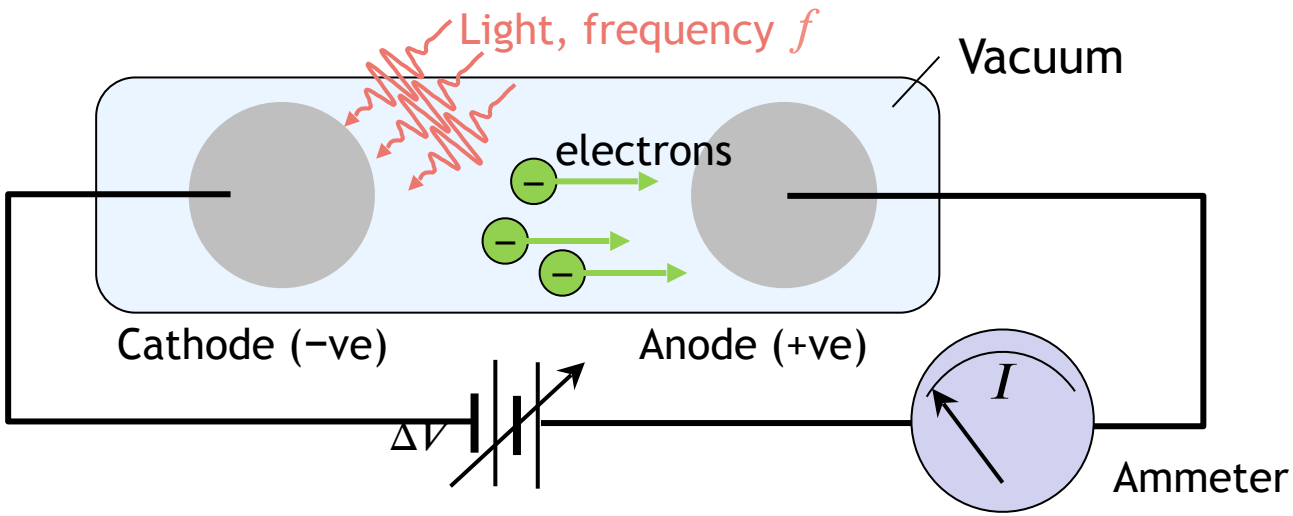


- Slope is constant
- y-intercept gives work function as a voltage, V_0

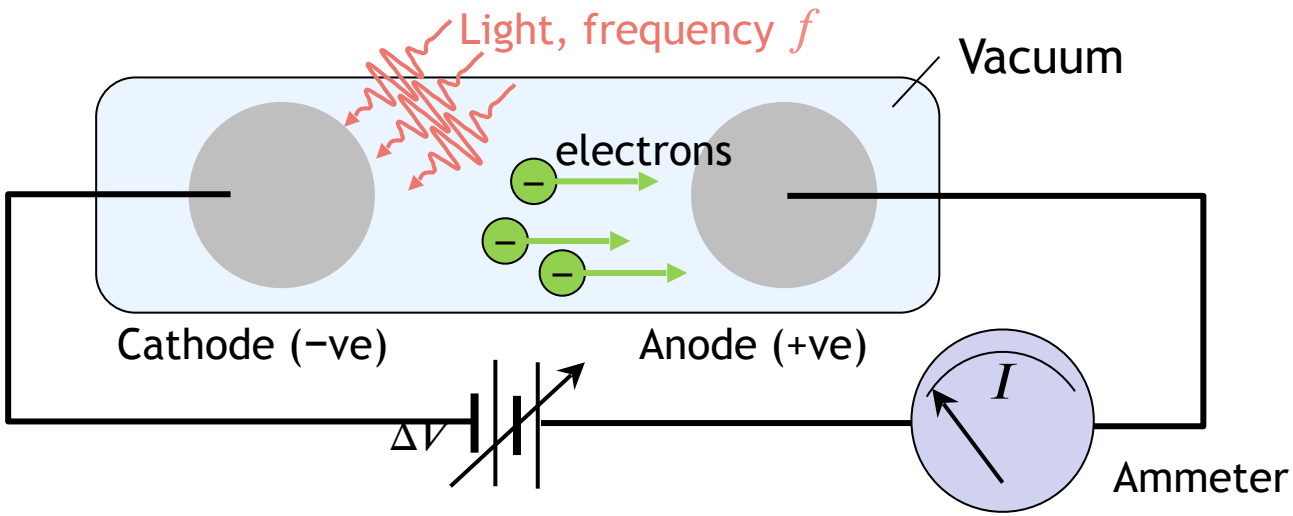
Photoelectric effect

1. Current is proportional to *intensity*



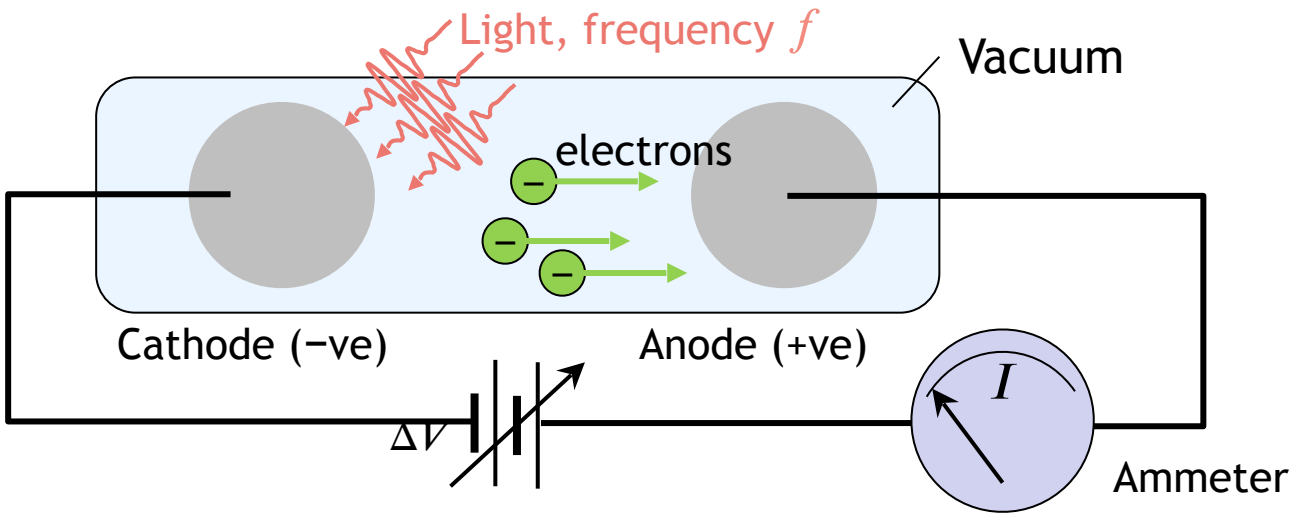
Photoelectric effect

1. Current is proportional to *intensity*
2. Current appears *immediately*



Photoelectric effect

1. Current is proportional to *intensity*
2. Current appears *immediately*
3. Current only if frequency is above some minimum *threshold*



Photoelectric effect

1. Current is proportional to *intensity*
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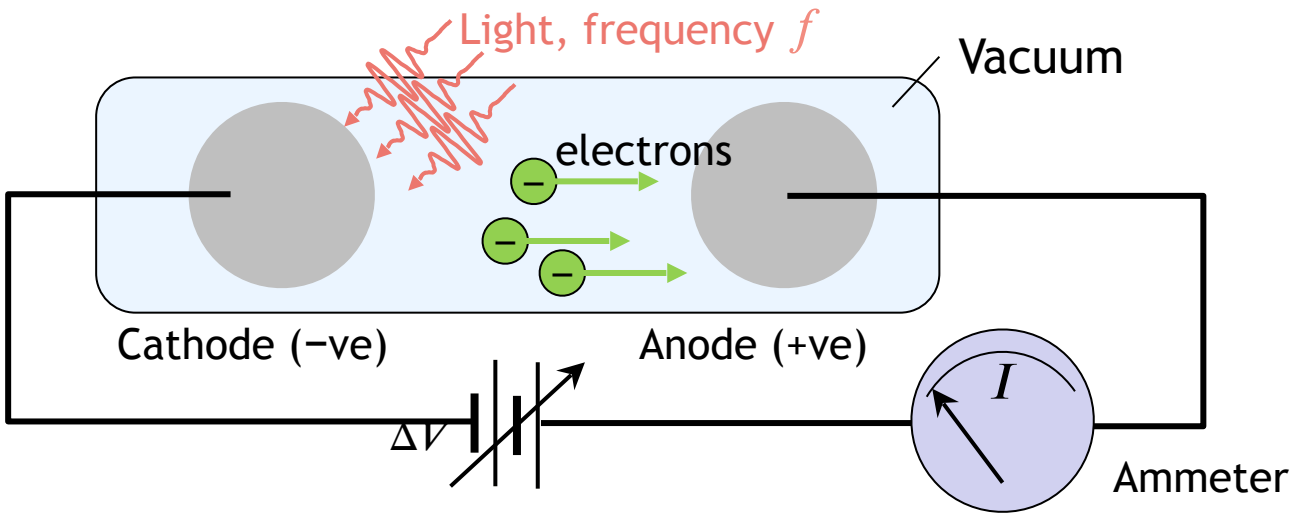
4. Threshold frequency depends on cathode metal



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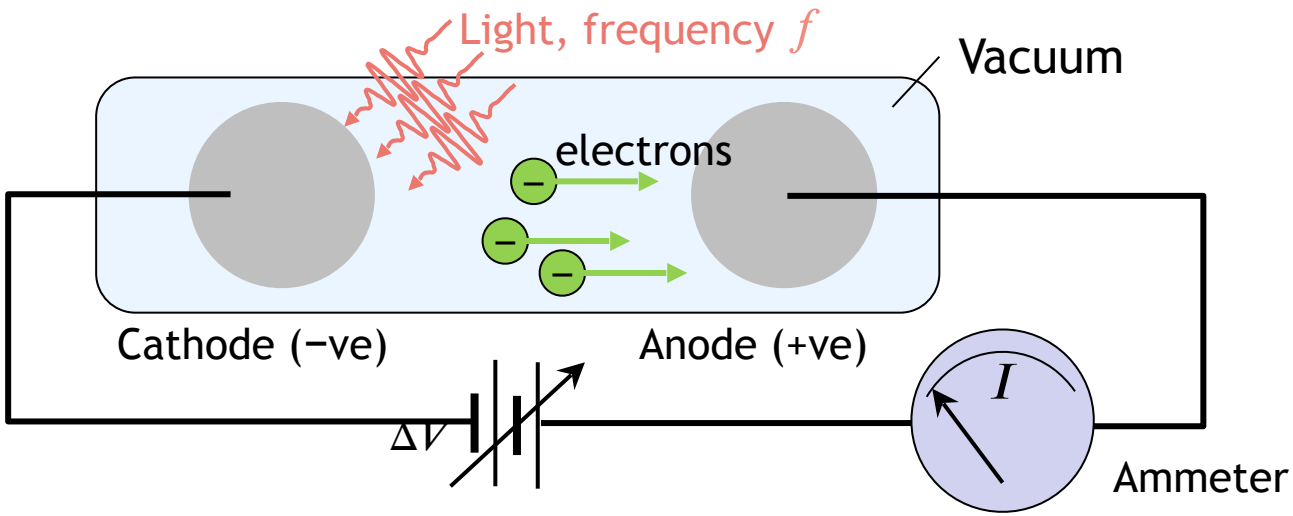
5. Current stopped by applying some minimum *stopping potential* V_{stop}



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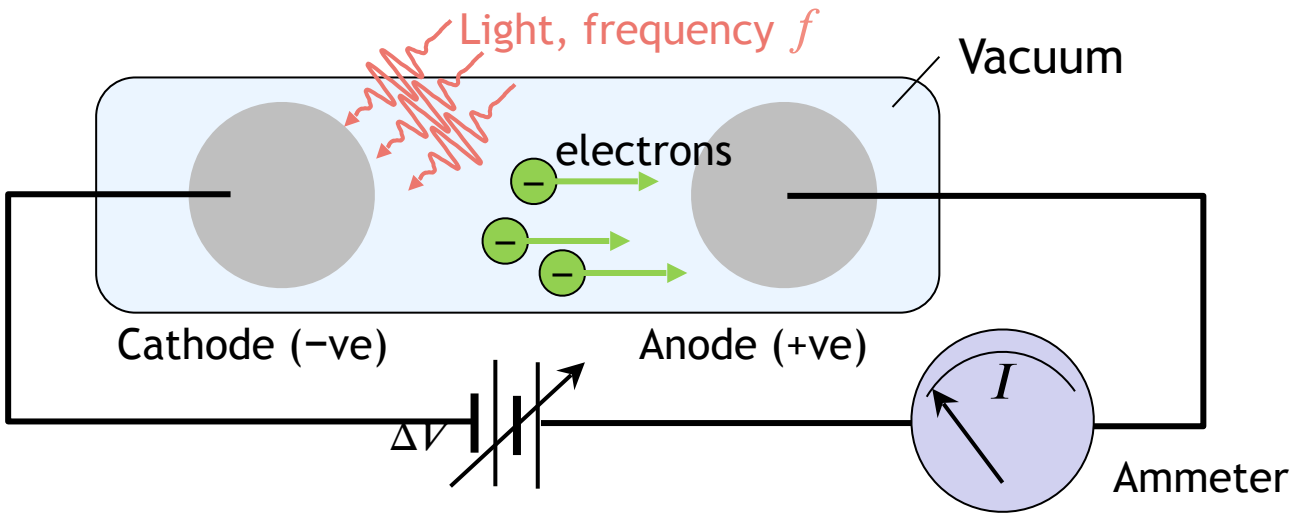
2. Current appears *immediately*
3. Current only if frequency is above some minimum *threshold*
4. Threshold frequency depends on cathode metal
5. Current stopped by applying some minimum *stopping potential* V_{stop}
6. Stopping potential depends on *frequency*

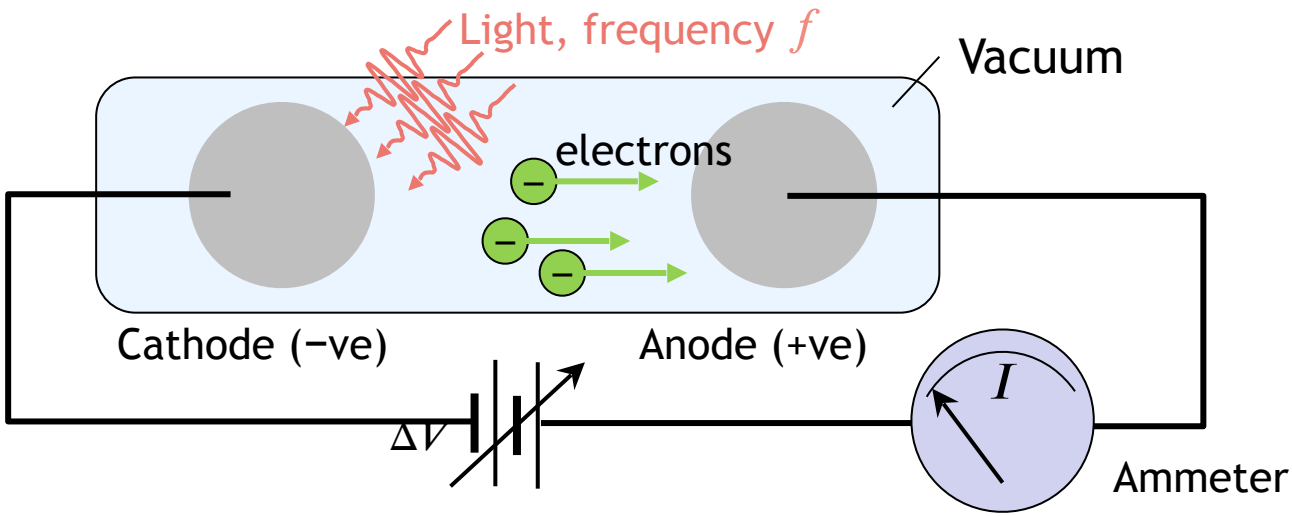


Classical photoelectric effect

Classical expectations

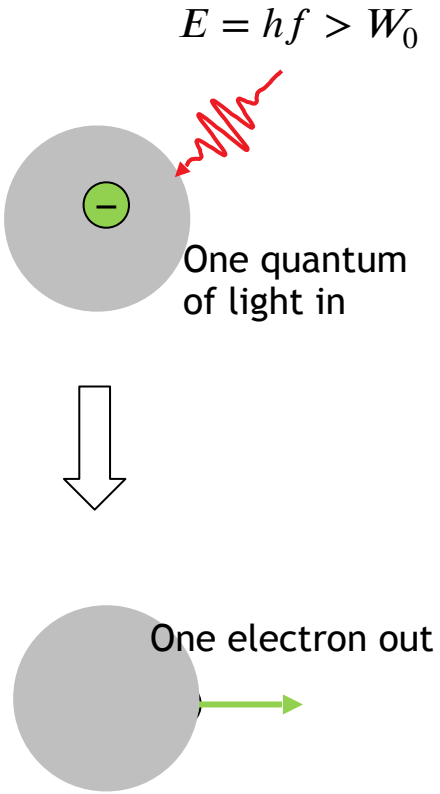
1. Electron energy increases with light intensity
2. Electron energy independent of light frequency
3. Time delay: material absorbs EM wave for some time before electron emitted

Measure electron energy by adjusting ΔV: reverse battery so $\Delta V < 0$, adjust until electrons stopped. Expect that to vary with intensity but not with f.



Photoelectric effect

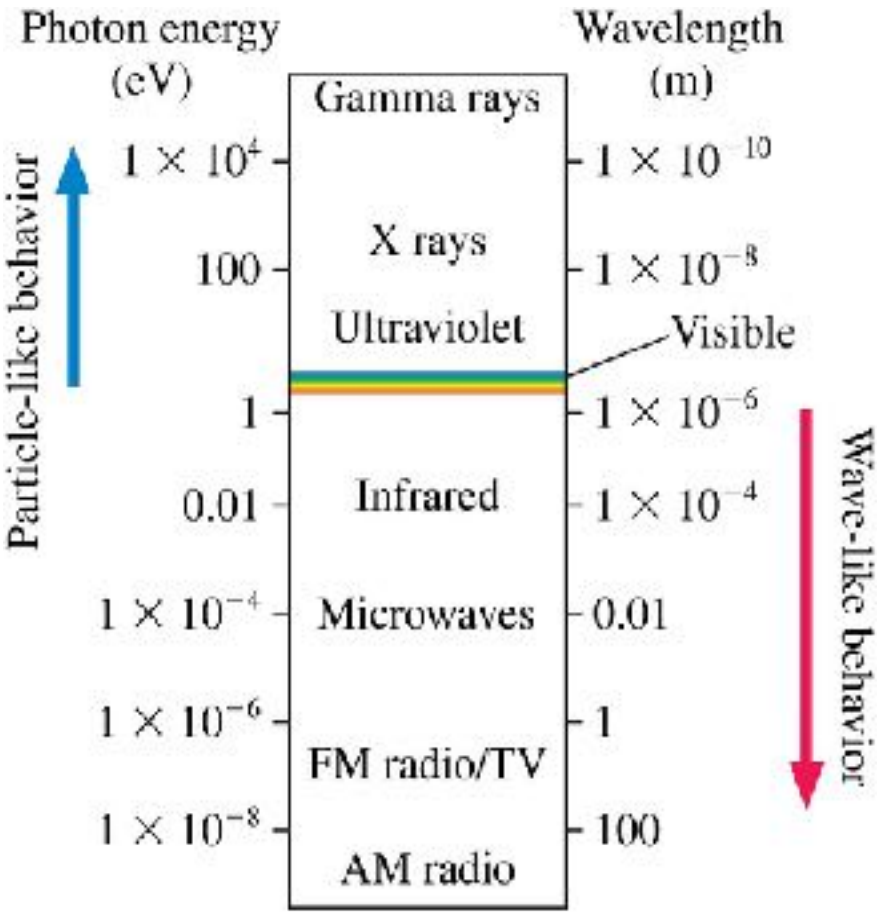
- EM radiation is absorbed in discrete packets (photons)
- Each packet has energy dependent on frequency $E = hf$
- Electrons are ***bound*** to cathode;
  requires certain amount of energy to release
- Electron released only if energy (frequency) of photon is high enough
- ***Binding energy*** of the electrons is called the ***Work Function*** W_0 *of the metal*
- We require $E = hf > W_0$
- That is, $f > f_0 = \dfrac{W_0}{h}$



Photons and Electromagnetic Radiation

- The wave-particle duality of Electromagnetic Radiation is well established.
- If you look for the wave-like properties of E/M radiation you'll find it.
- If you look for the particle-like properties of E/M radiation you'll find them.
- We'll use this to examine Magnetic Resonance Imaging (MRI)

(Our most useful form of non-invasive imagining)



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Work Function

Red light is incident on a piece of metal in a vacuum. No electrons are released. The wavelength is reduced until the light is yellow, when electrons are released. If the metal is replaced with a metal having a higher work function, which colour of light would have the best chance of releasing electrons from the metal?

A. Blue

B. Red

C. Yellow

D. We need to know more

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Photon Intensity

Beams of red and blue light have equal intensities. Which of the following statements is true?

A. There are more photons per second in the blue beam

B. There are more photons per second in the red beam

C. Both beams contain the same number of photons per second

D. The number of photons per second is not related to intensity.

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Photon Energy

Which of the following is necessarily true?

A. Red light has more energy than violet light.

B. Violet light has more energy than red light.

C. A single photon of red light has more energy than a single photon of violet light.

D. A single photon of violet light has more energy than a single photon of red light

E. (A and C)

F. (B and D)

G. None of the above

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Energetic electrons

If a photon of energy E ejects electrons from a metal with kinetic energy KE, then a photon with energy $E/2$

A. Will eject electrons with kinetic energy KE\/2

B. Will eject electrons with an energy greater than KE\/2

C. Will eject electrons with an energy less than KE\/2

D. Might not eject any electrons.

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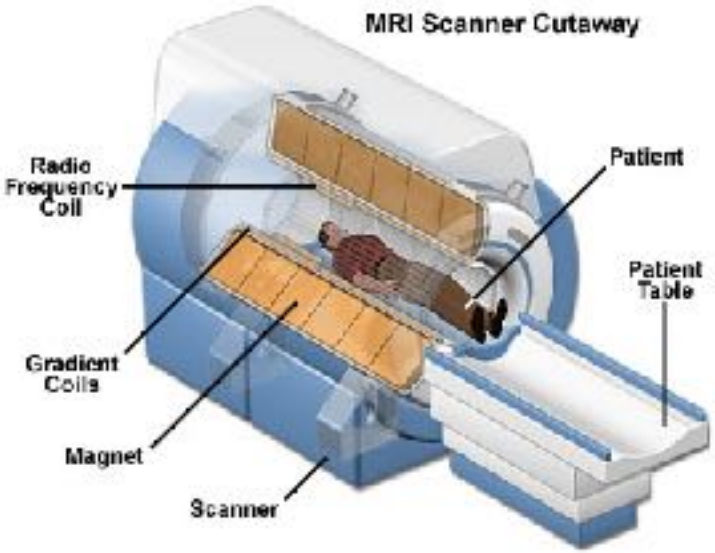
31.9 Nuclear magnetic resonance and MRI



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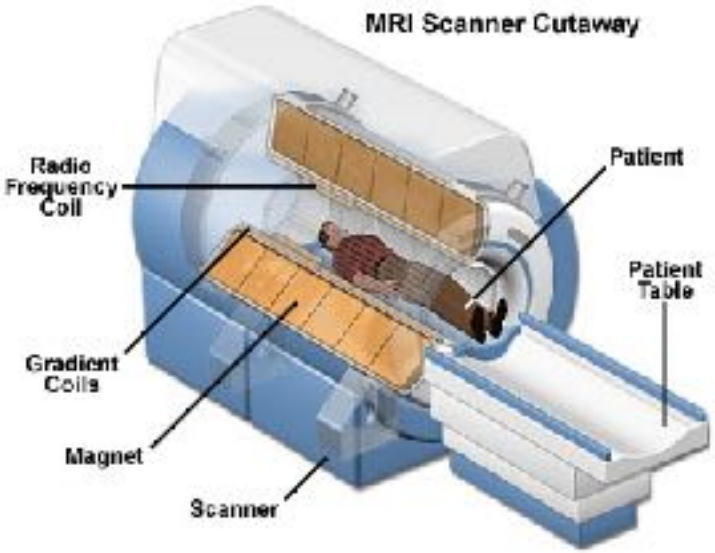
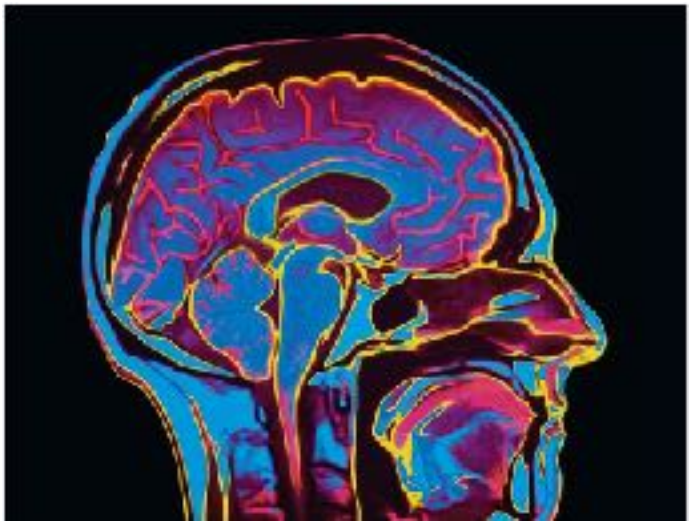




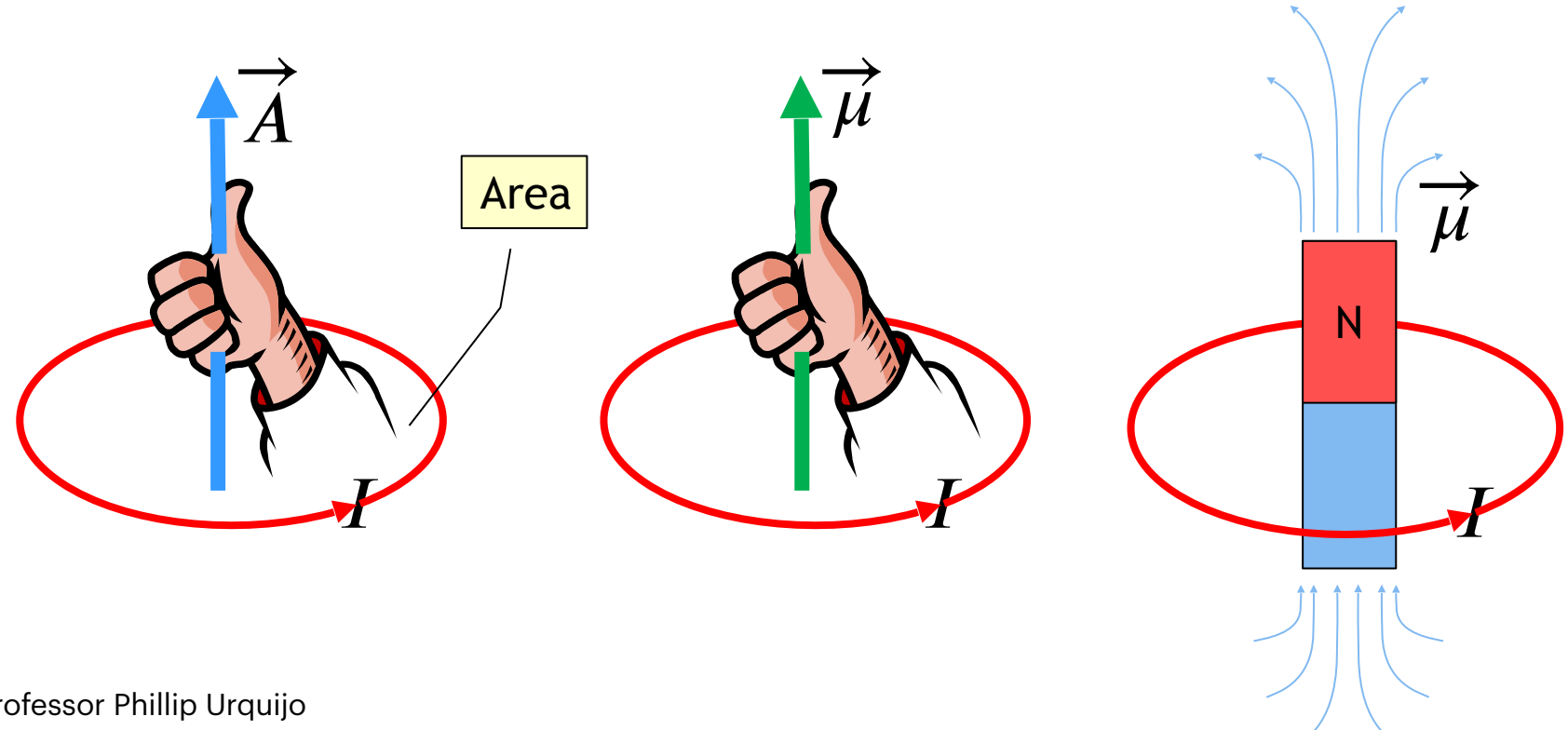
20.9 Magnetic Dipoles

- Loop of current = “tiny bar magnet” = ***magnetic dipole***

- Magnetic dipoles feel torque that aligns them to an external magnetic field

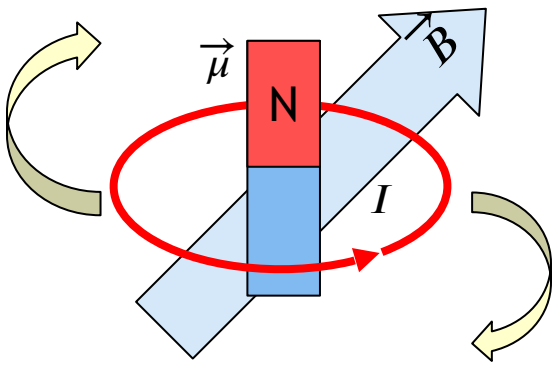
Magnetic dipole moment $\vec{\mu} = I\vec{A}$

$\vec{\mu}$ is the magnetic dipole moment
Units: Joules/Tesla, J/T
$\vec{A}$ is the area (vector)
I is the current in the loop



Magnetic dipoles in an external magnetic field

- Like compass needles, magnetic dipoles twist to align with external field



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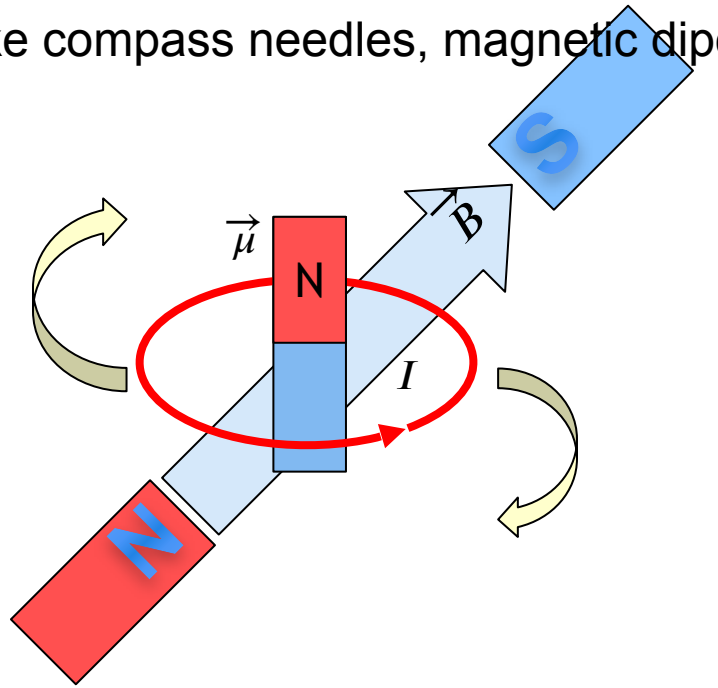

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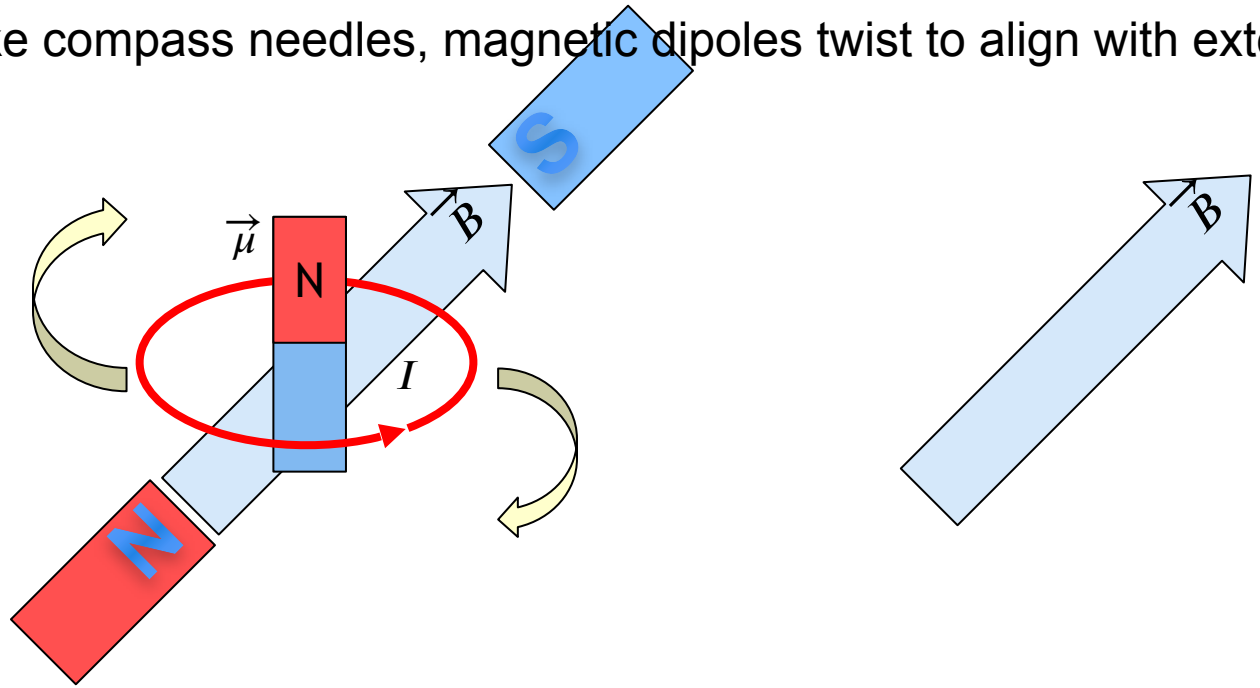

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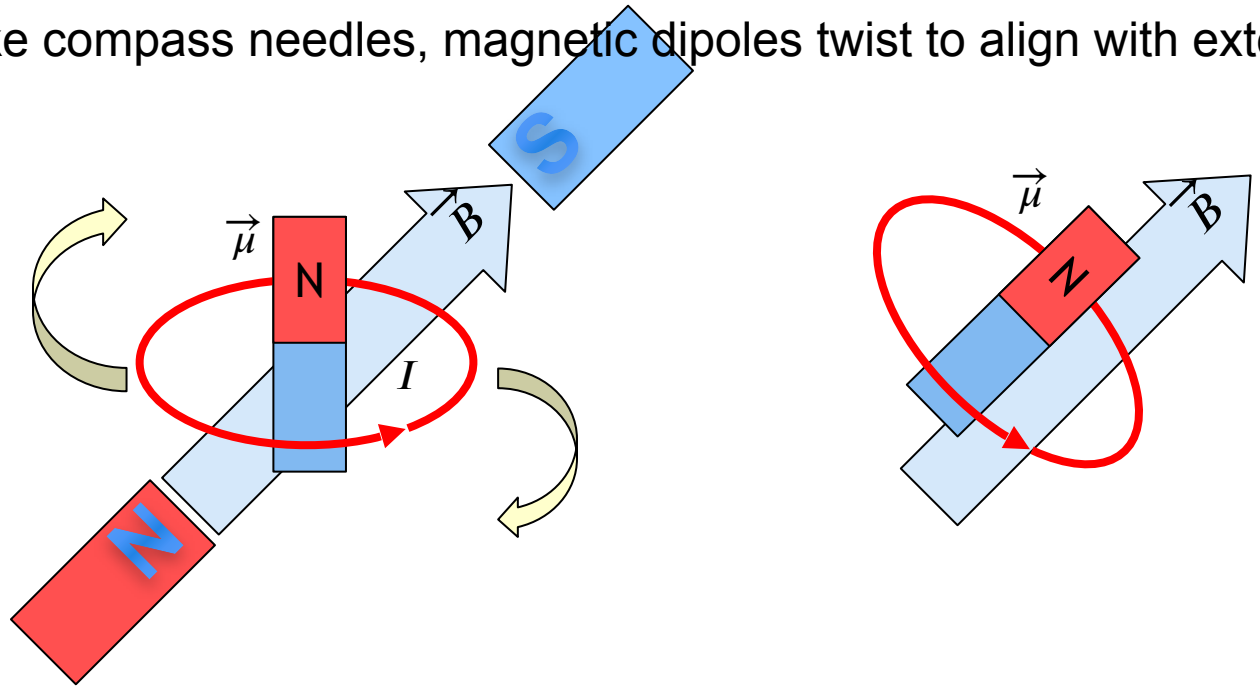

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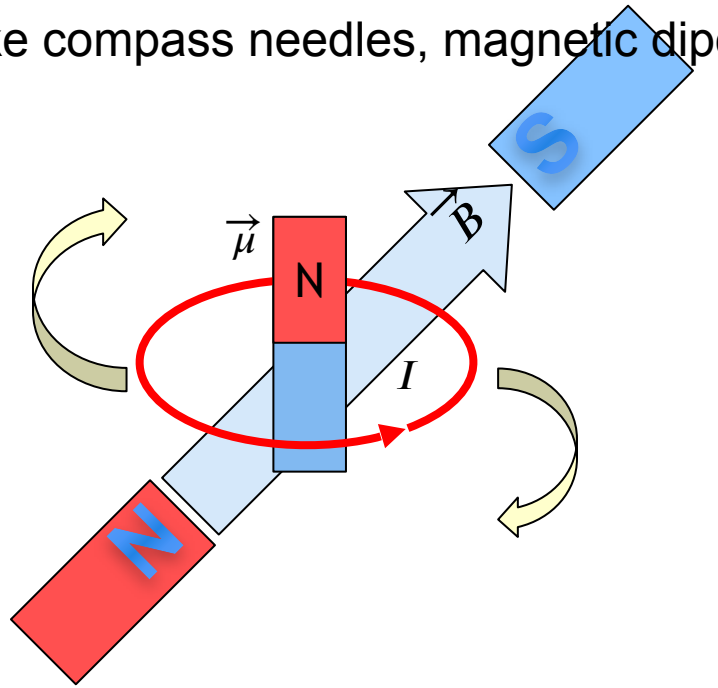

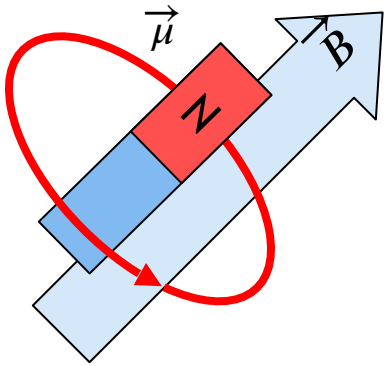

$$U = -\vec{\mu} \cdot \vec{B} = -\mu B \cos 180^\circ = +\mu B$$

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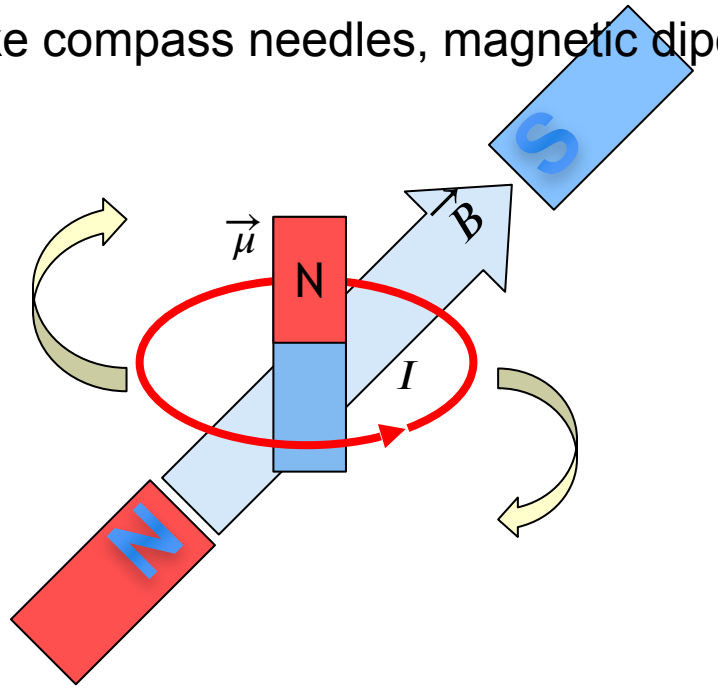

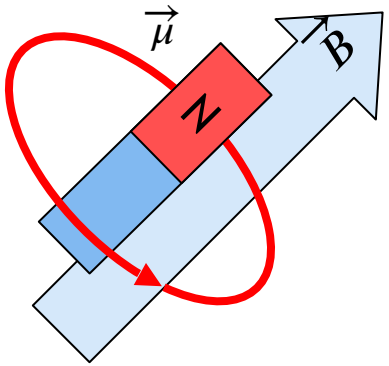

- Non-aligned = potential energy
- Aligned = lower energy

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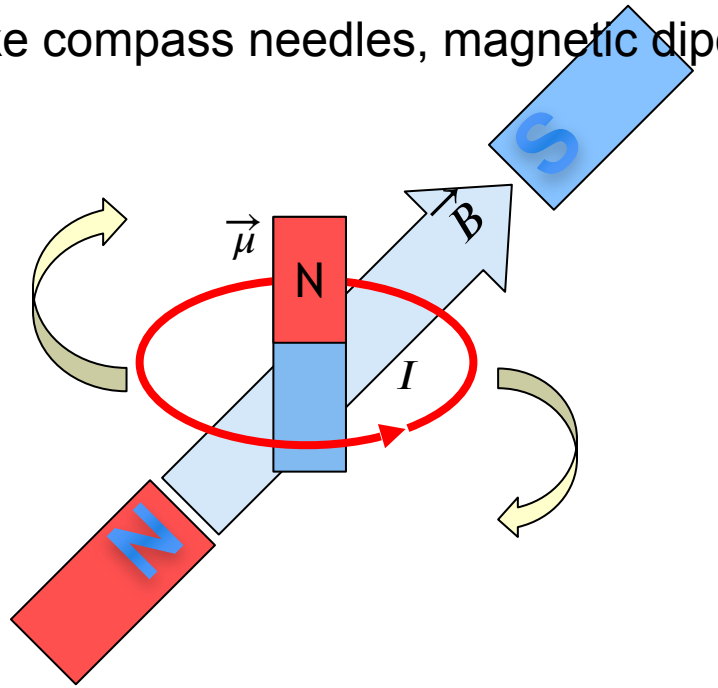

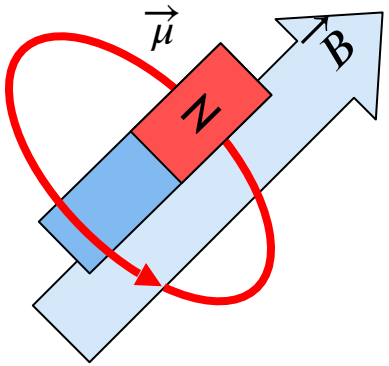

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- Energy difference

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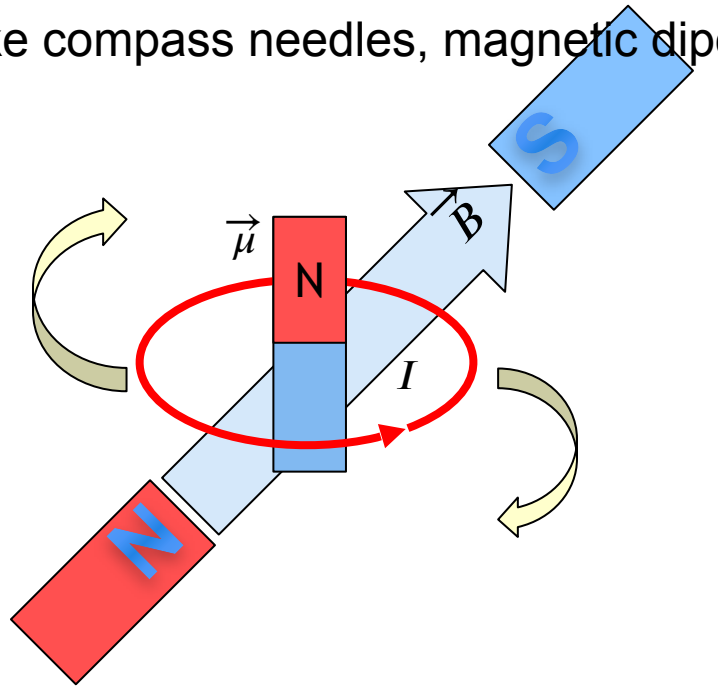

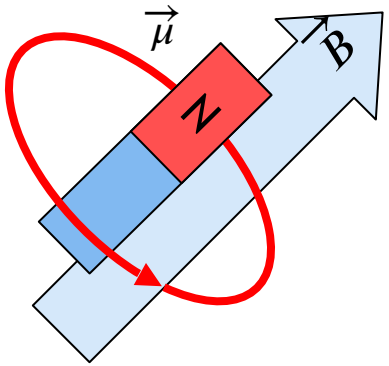

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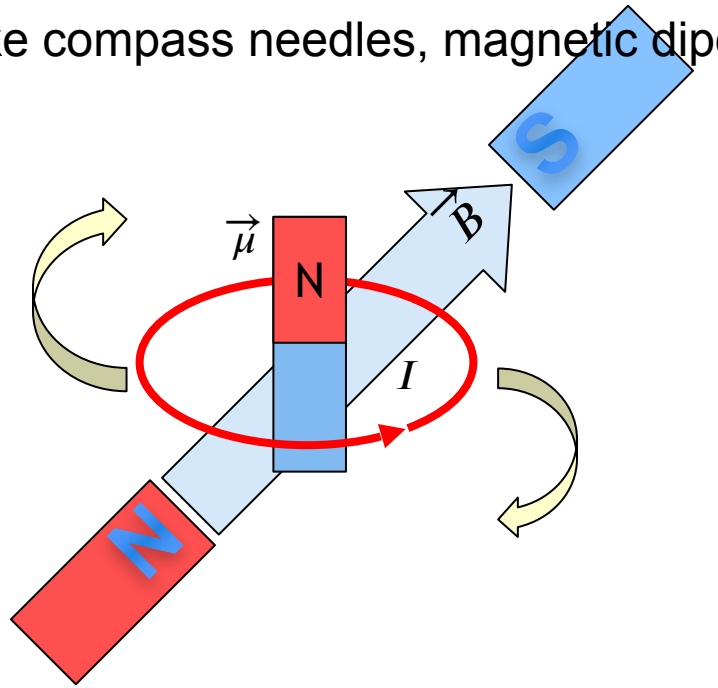

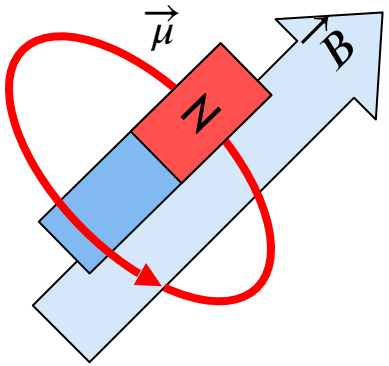

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- Energy difference $\Delta U = 2\mu B$

- *Note: "aligned" means external field lines and field lines of dipole are aligned*

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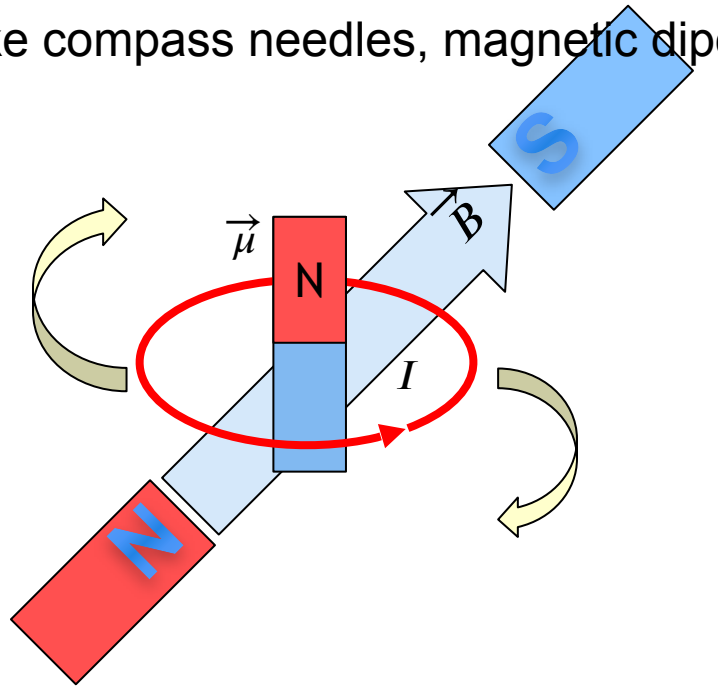

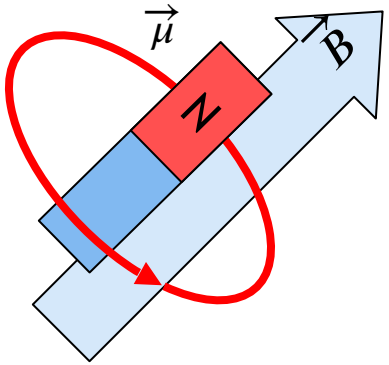

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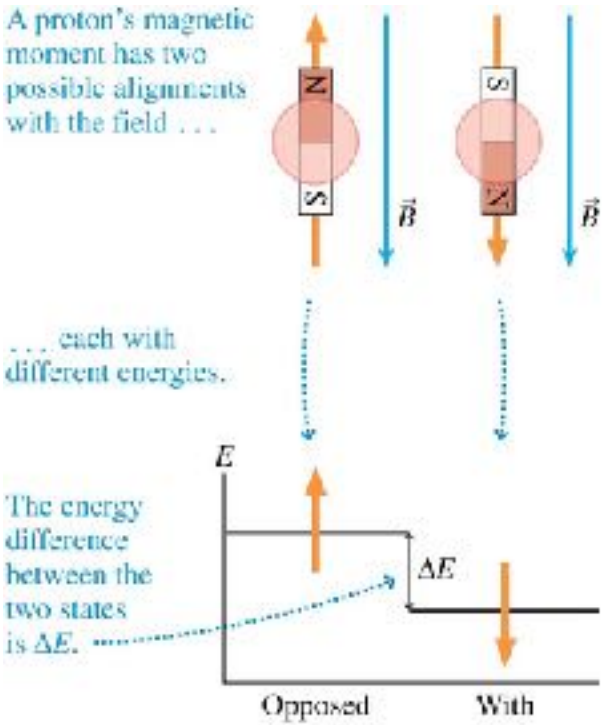
D Na-5 Nuclear Magnetic Resonance

Protons (and electrons, neutrons) are little tiny magnets!

- Protons have a magnetic dipole moment
  - Small bar magnet $\mu = 5\ \mathrm{J/T}$
  - Earth $\mu = 8 \times\ 10^{22}\ \mathrm{J/T}$
  - Proton $\mu_p = 1.41 \times 10^{-26}\ \mathrm{J/T}$
  - Electron $\mu_e = 657\ \mu_p$ (FRIDGE MAGNETS!)



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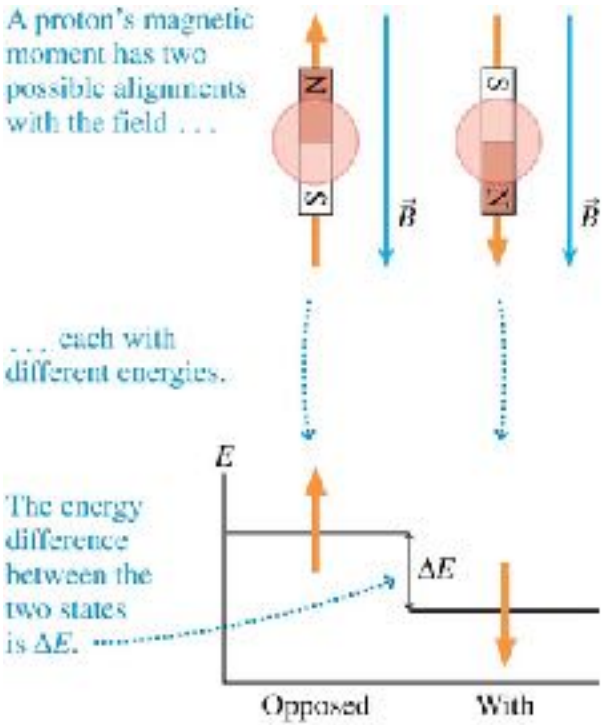
- Proton in external magnetic field of 1 T



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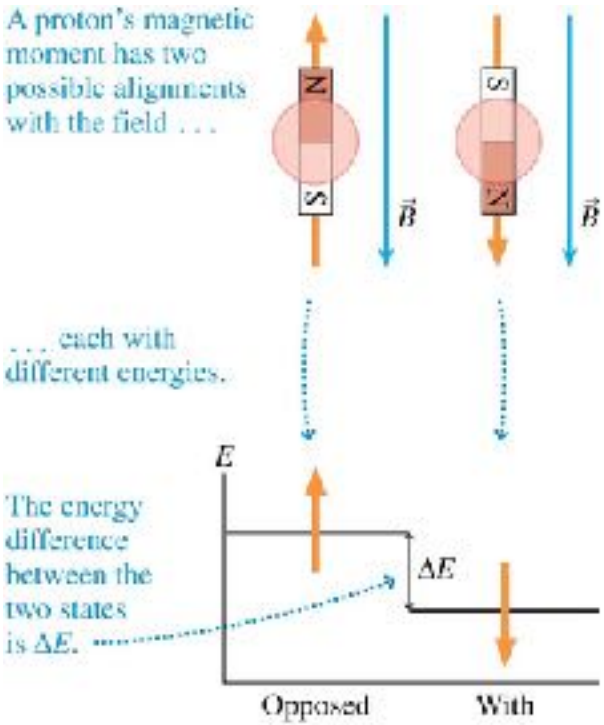
- Proton in external magnetic field of 1 T

$$\Delta U = 2\mu B = 2.8 \times 10^{-26}\ \mathrm{J/T}$$



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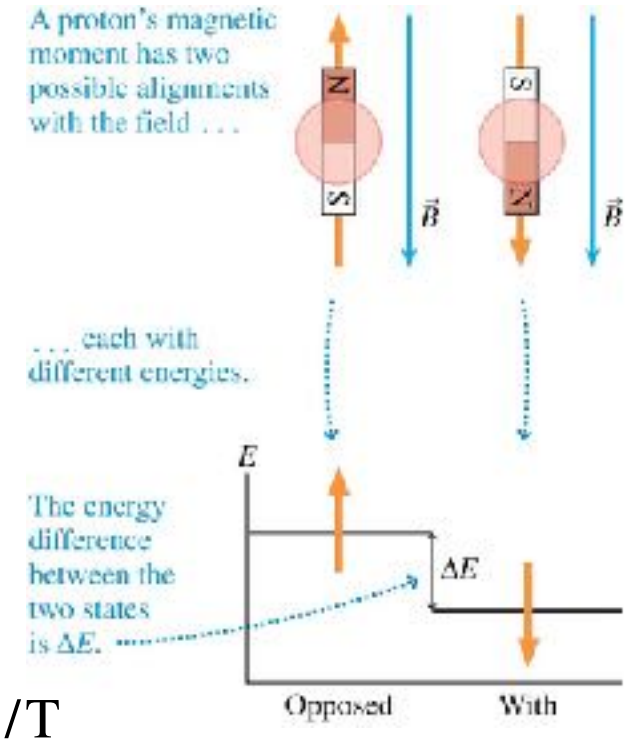

- Proton in external magnetic field of 1 T

$$\Delta U = 2\mu B = 2.8 \times 10^{-26}\ \mathrm{J/T}$$

- Corresponds to EM frequency of…

$$E = hf \rightarrow f = \frac{\Delta U}{h} = \frac{2.8 \times 10^{-26}}{6.63 \times 10^{-34}} = 42\ \mathrm{MHz/T}$$

Protons (and electrons, neutrons) are little tiny magnets!

- Protons have a magnetic dipole moment
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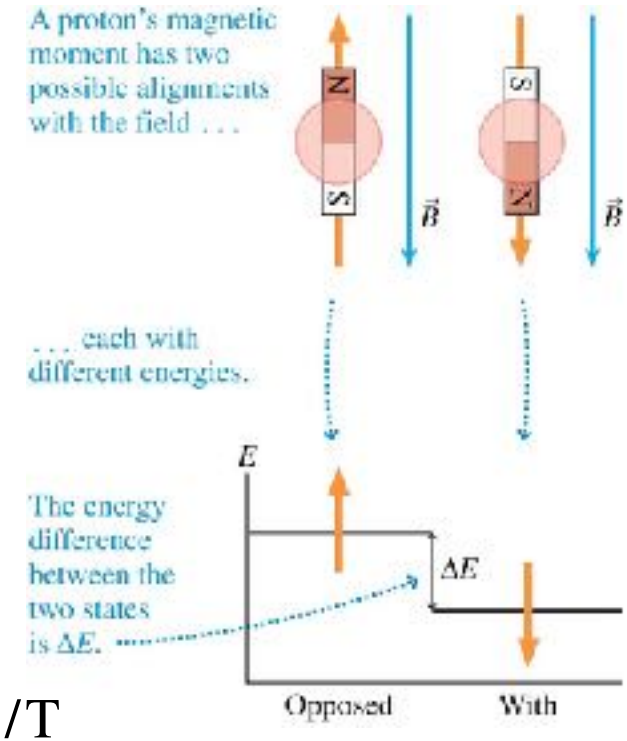

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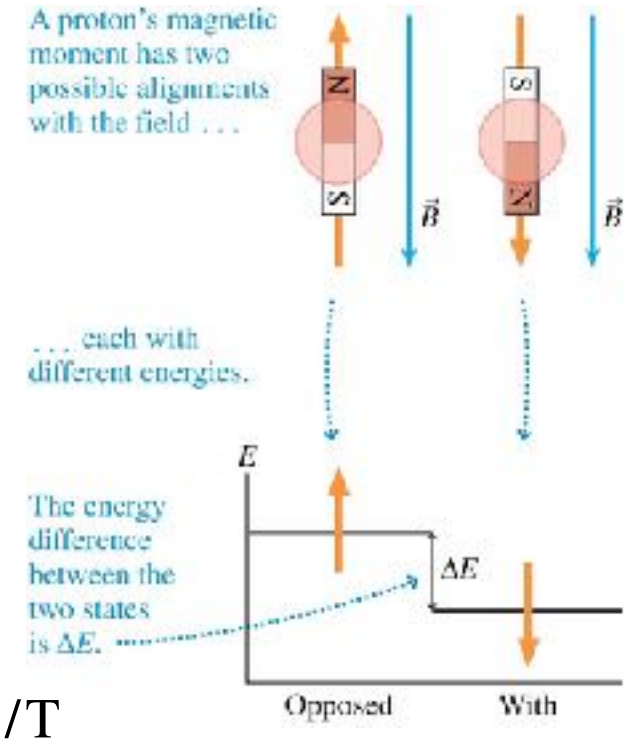

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- A *radio* wave can flip a proton in a magnetic field upside down!

Sample exam-style question

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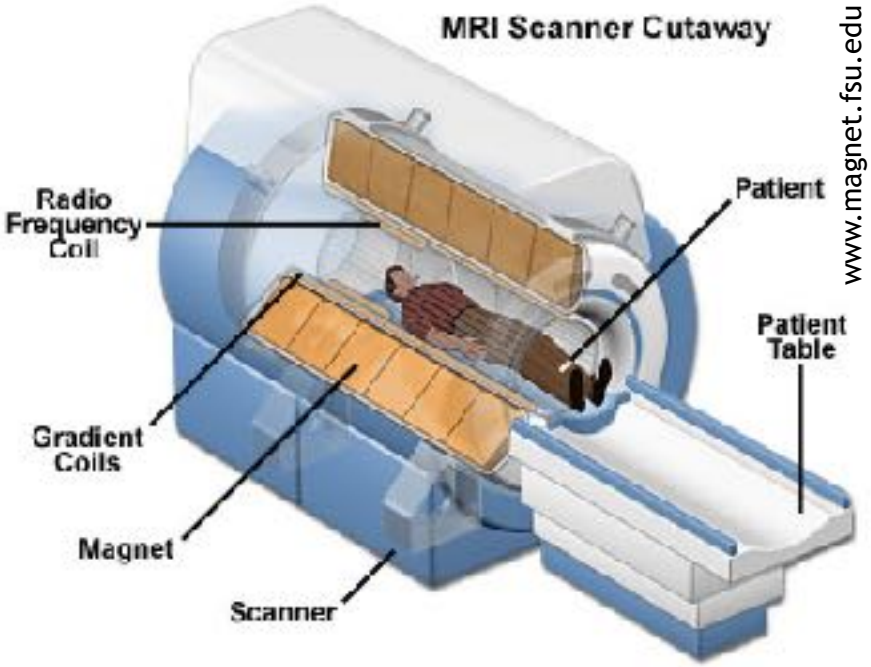
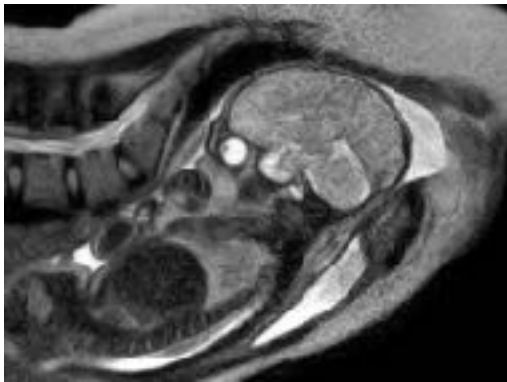
c) Ratio is about $1\mathrm{eV}/2 \times 10^{-13}\mathrm{eV}$ = 5 trillion to one.

Magnetic Resonance Imaging (MRI)

- Patient in a strong magnetic field
- Radio tuned to $f = 2\mu B/h$ resonance
- Flipped protons fall back; emitting radio
- *B* varies: different regions, different frequency
- Local proton density (chemistry) affects emitted radio
- Local chemistry also affects local *B*; “chemical shift”



www.magnet.fsu.edu



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- Magnetic dipole moments $\vec{\mu} = I\vec{A}$
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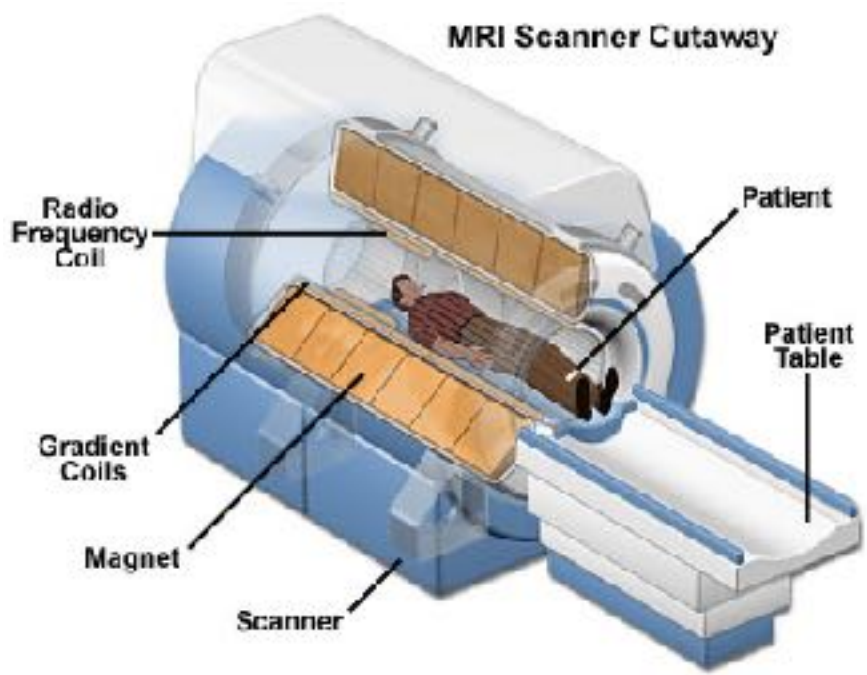

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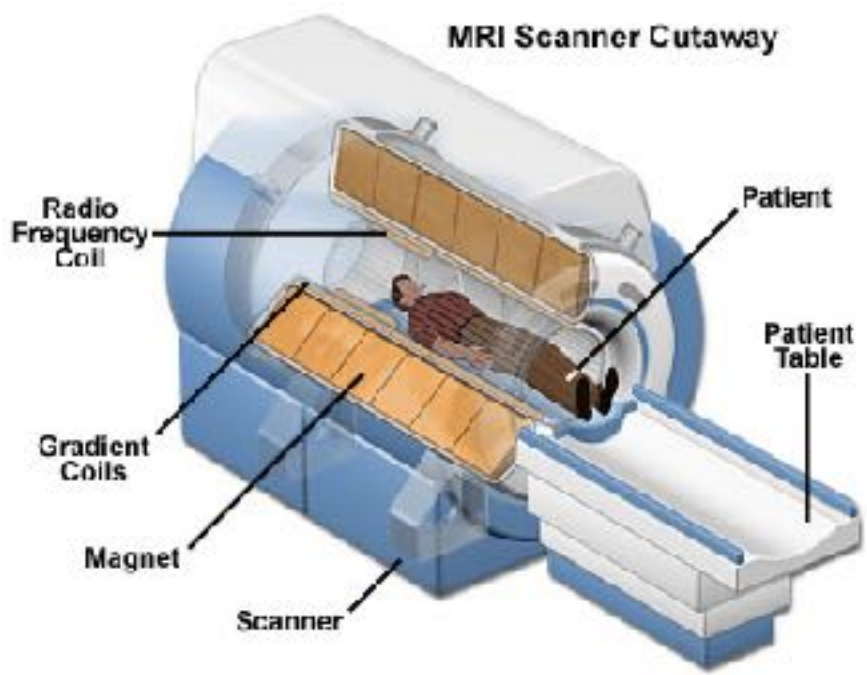
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